From Human Markets to Machine Markets:

Feedback Stability, Auditable Memory, and Financial Constraints

Pratik Niroula[*] and **Prashant Niroula**[†]

October 6, 2026 — revision candidate

Abstract

Aggregate price–flow data reveal model-dependent feedback diagnostics rather than trader identities. In a reset linear cohort model, symmetric near-coincident rates require bounded-power probing budgets of at least order δ^{-4}; a separate scalar Gaussian experiment yields a finite-window abstention layer. Reanalysis of recorded Coinbase data shows substantial sensitivity to flow timing and price filtering, while covariance restrictions are rejected. Raw reconstruction of 768 Binance asset-days confirms smaller feedback under observed-flow timing; fee contrasts and ordering-sensitive venue decompositions do not identify causal learning or transmission. Finally, a declared loss operator and shock set yield an exact budget radius; fixed-spectrum and funded-dealer simulations distinguish transient magnification from persistent-flow accumulation. Independent replication and policy validation remain open.

Keywords: Adaptive learning, algorithmic trading, financial stability, market microstructure, system identification, non-normal dynamics, Kreiss constant.
JEL Classification Codes: G01, G10, G14, C32, C58, D83.

[*]Corresponding author email: `pratik.niroula@alumni.harvard.edu`. Repository address reported in the original manuscript (remote contents not verified in this revision): `github.com/pratikniroula/machine-markets` (Revision baseline: `451fd803c516`. This candidate has not been independently replicated).

[†]Email: `prashant.niroula@alumni.harvard.edu`.

1 Introduction

Electronic trading creates a feedback channel when price changes affect subsequent orders and those orders affect later prices. The existence of this channel does not identify its participant-level source: aggregate flow can reflect adaptive algorithms, order splitting, liquidity provision, or common information. Nor does decay of price disturbances establish that a dealer's loss budget will remain intact. This paper separates these identification and financial-risk questions.

Our central question is: *what can observable aggregate data establish about feedback, and what additional assumptions are required to turn dynamic stability into a financially meaningful loss bound?* We combine a linear adaptive-memory model with an audit of observable-state definitions and timing, followed by controlled loss-exposure illustrations. The empirical analysis is deliberately reduced-form. It does not identify artificial-intelligence trading, individual learning rules, or a causal market-maker response from anonymous records.

Conditional identification results. Within a reset linear cohort model with known price-impact and anchor parameters, price responses identify merged rate–weight measures rather than the number or identity of cohorts sharing a rate. For a uniformly stable compact family of symmetric split alternatives, bounded-power deterministic probes with Gaussian observation noise face a separation lower bound of order δ^{-4}. This is a lower bound for the stated experiment, not an attained universal rate for all passive market data. A separate scalar Gaussian price-surveillance experiment gives an explicit finite-window abstention bound near its unit-root boundary. The observation experiment matters: the scalar result is not asserted for noiselessly observed latent cohort states.

Measurement sensitivity. The primary empirical reanalysis uses recorded Coinbase BTC, ETH, and ADA book snapshots and signed executed flow from April 7–19, 2021. A 300-second EMA price deviation is estimated on 10-second bins with only end-bin observed flow. Mean reduced-form feedback is approximately 0.00194, 0.00098, and 0.00062, respectively. Forward-bin flow alignment produces larger estimates, while raw-return specifications have feedback intervals spanning zero. These comparisons establish that measurement and timing materially affect the feedback diagnostic. They do not establish an algorithmic learning mechanism.

The preserved Binance panel supplies a secondary analysis of 768 saved asset-day estimates, derived from trade-price klines rather than historical full order books. Its legacy forward-flow estimator has mean $\hat{Q} \simeq 0.00815$. The initial cache audit lacked those raw inputs. Section F now restores all 768 original asset-days from checksum-verified official

archives, reproduces the saved core estimates, and reports corrected observed-flow diagnostics separately. Original download-time hashes were not preserved. The legacy July 2022 fee contrast has a date-clustered effect interval spanning both signs and fails the inherited equivalence reference. Corrected observed-time results in Section F have a different scale; a wide inherited margin cannot establish economic equivalence. The contradictory old joint-venue numbers and unsupported event-study inference are withdrawn. A subsequent independent reconstruction of the original April overlap supplies a joint VAR and ordering-sensitive predictive decompositions in Section E.

Financially meaningful stability. For fixed economically specified state coordinates, loss operator C, and initial-shock set, the exact loss-budget radius is $B/\sup_{t\geq 0}\|CJ^t\|_2$. The underlying transient-growth mechanism is established in operator theory; our financial use is the explicit mapping from a declared disturbance set to a declared budget. A fixed-spectrum numerical illustration separates transient magnification from eigenvalue decay. A distinct funded-dealer simulation shows persistent-flow accumulation and retains an exogenous benchmark matched on long-run variance. Simulation outcomes are conditional on their accounting and calibration, not measurements of actual dealer insolvency.

1.1 Related Literature and Contribution Boundaries

Classical microstructure studies price formation, information and order flow (Kyle, 1985; Glosten and Milgrom, 1985; Hasbrouck, 1991). Order splitting and book imbalance can create persistent relations without identifying an adaptive trader (Bouchaud et al., 2004; Cont, Kukanov, and Stoikov, 2014). Funding constraints and inventory exposure connect market liquidity to intermediary fragility (Brunnermeier and Pedersen, 2009; Adrian and Shin, 2010; Greenwood, Landier, and Thesmar, 2015).

The realization and transient-growth tools used here build on existing control and numerical analysis (Gilbert, 1963; Kreiss, 1962; Spijker, 1991; Trefethen and Embree, 2005). Non-normal financial-network amplification is not a first-use claim: Wang, Sornette, Wu and Lera's 2024 working paper studies evolving influence networks and subcritical financial instabilities (Wang et al., 2024). Singular mixture models also have assumption-dependent identification rates (Manole and Ho, 2022); our symmetric probing experiment is a specific dynamical testing problem rather than a universal new estimation rate. Our scope differs by specifying an output loss budget, a cohort-response observation experiment, and sensitivity of measured price–flow feedback to timing and filtering. We do not claim novelty for the general distinction between asymptotic decay and transient growth, or an independently certified priority for the full framework.

1.2 Evidence Organization

Section 2 states the conditional model results. Section 3 defines actual measurements and their timing. Section 4 reports fresh observable-state diagnostics and clearly separated saved-panel evidence. Section 5 examines dimensional stress regression and model-based loss exposure. Section 6 states attribution and replication limits. Section 7 discusses conditional surveillance implications and concludes.

2 Theoretical Model and Identification Boundaries

2.1 Microfoundations of Closed-Loop Price Formation

We formalize an electronic financial market sampled at discrete uniform time intervals $t \in \{0, 1, 2, \dots\}$, where time step Δ represents the operational sampling frequency (e.g., 1 second). Let p_t denote the log mid-price of a traded asset, and let p_t^* denote the unobserved fundamental value, which evolves as a martingale:

$$p_{t+1}^* = p_t^* + \xi_{t+1}, \qquad \xi_{t+1} \sim \text{i.i.d.}(0, \sigma_\xi^2). \tag{1}$$

We define the pricing error (displacement from fundamental) as $x_t \equiv p_t - p_t^*$.

Dealer Pricing and Kyle Auction Microfoundation. Following the continuous auction framework of Kyle (1985) adapted to discrete quote adjustments (Hasbrouck, 1991; Bouchaud et al., 2004), a competitive, risk-neutral market maker sets quotes to absorb aggregate order flow. Total incoming order flow at time t consists of two components: aggregate demand u_t submitted by algorithmic trading agents, and exogenous liquidity/hedging noise $\varepsilon_{t+1} \sim \text{i.i.d.}(0, \sigma_\varepsilon^2)$. In addition, fundamentalist arbitrageurs submit mean-reverting demand $-\varphi x_t$ whenever prices deviate from fundamentals.

The market maker updates the transaction price according to linear price impact $\kappa > 0$:

$$p_{t+1} - p_t = \kappa \left(u_t - \varphi x_t + \varepsilon_{t+1} \right). \tag{2}$$

Subtracting fundamental increment ξ_{t+1} from both sides yields the pricing error recursion:

$$x_{t+1} = (1 - \kappa\varphi) x_t + \kappa u_t + (\kappa\varepsilon_{t+1} - \xi_{t+1}). \tag{3}$$

Defining the direct price persistence parameter $a \equiv 1 - \kappa\varphi \in [0, 1)$ and rescaling the combined

disturbance innovation as $w_{t+1} \equiv \kappa\varepsilon_{t+1} - \xi_{t+1}$, we obtain the foundational pricing equation:

$$x_{t+1} = ax_t + \kappa u_t + w_{t+1}, \qquad w_{t+1} \sim \text{i.i.d.}(0, \sigma_w^2). \tag{4}$$

When fundamentalist arbitrage is strong ($\varphi \to 1/\kappa$), $a \to 0$ and prices immediately reflect fundamentals. When liquidity is thin and arbitrage capital is limited, a approaches unity, reflecting high price persistence.

Algorithmic Learning Microfoundations. We consider K heterogeneous cohorts of algorithmic traders. Each cohort $i \in \{1, \ldots, K\}$ maintains an internal state variable $y_{i,t} \in \mathbb{R}$ representing its estimate of price momentum, trend, or fundamental mispricing. We show that two stylized signal-update paradigms—steady-state Kalman filtering and regularized Online Gradient Descent—both reduce locally to the identical linear adaptive recursion.

Proposition 2.1 (Steady-State Kalman Filtering Yields Exponential Memory)**.** *As an independent microfoundation for exponential-memory signals, suppose an algorithmic execution desk tracks a latent local trend $v_t \in \mathbb{R}$ following a random walk $v_{t+1} = v_t + \zeta_{t+1}$ with innovation variance $\sigma_\zeta^2 > 0$, observing noisy price return increments $r_t = v_t + \nu_t$ with observation noise $\sigma_\nu^2 > 0$. Assume a Gaussian initial state and mutually independent, zero-mean Gaussian innovation and observation-noise sequences, independent of the initial state. In the steady-state covariance regime, the minimum mean-squared error (MMSE) estimator follows:*

$$\hat{v}_{t+1} = (1 - \eta)\hat{v}_t + \eta r_t, \tag{5}$$

where the steady-state Kalman gain $\eta \in (0, 1)$ is the unique positive root of the algebraic Riccati equation:

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2}. \tag{6}$$

Proof. The scalar state-space model is $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$. Let P_∞ denote the steady-state prior error variance. The Riccati equation is $P_\infty = P_\infty - \frac{P_\infty^2}{P_\infty + \sigma_\nu^2} + \sigma_\zeta^2$, which simplifies to $P_\infty^2 - \sigma_\zeta^2 P_\infty - \sigma_\zeta^2\sigma_\nu^2 = 0$. The unique positive root is $P_\infty = \frac{\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2}$. The steady-state Kalman gain is $\eta = \frac{P_\infty}{P_\infty + \sigma_\nu^2} \in (0, 1)$. Substituting into the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ yields (5). □

Proposition 2.2 (Regularized Online Gradient Descent Yields Exponential Memory)**.** *Consider an automated trading agent updating policy weights $\theta_t \in \mathbb{R}$ via Online Gradient Descent (OGD) with L_2 weight regularization $\gamma \in (0, 1)$ to minimize a quadratic prediction loss*

$\mathcal{L}(\theta_t; x_t) = \frac{1}{2}(x_t - \theta_t)^\top H(x_t - \theta_t)$:

$$\theta_{t+1} = (1-\gamma)\theta_t - \alpha \nabla_\theta \mathcal{L}(\theta_t; x_t), \tag{7}$$

where $H > 0$ *and* $\alpha > 0$ *is the learning step size. If the effective adaptation rate* $\eta \equiv \gamma + \alpha H \in (0,1)$ *and we define gain* $q \equiv \frac{\alpha H}{\gamma + \alpha H}$*, the parameter recursion is mathematically identical to:*

$$\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t. \tag{8}$$

Proof. Differentiating the loss, $\nabla_\theta \mathcal{L} = -H(x_t - \theta_t)$. The parameter update becomes: $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1 - (\gamma + \alpha H))\theta_t + \alpha H x_t$. Substituting $\eta = \gamma + \alpha H$ and $q = \frac{\alpha H}{\eta}$ yields $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$. □

Propositions 2.1 and 2.2 demonstrate that exponential memory decay is not an ad hoc modeling assumption, but rather a common local linear reduction of standard optimal filtering and regularized gradient optimization.

The Joint Closed-Loop Dynamical System. Let cohort i submit demand proportional to its state, $u_{i,t} = q_i y_{i,t}$, where $q_i \in \mathbb{R}$ reflects aggressive capital allocation ($q_i > 0$ represents trend-following / momentum; $q_i < 0$ represents mean-reversion). Total algorithmic demand is $u_t = \sum_{i=1}^K q_i y_{i,t}$. Each cohort updates its internal state according to:

$$y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i x_{t+1}, \qquad \eta_i \in (0,1). \tag{9}$$

Substituting the market price formation equation (4) into the update rule (9), we obtain:

$$y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i \left(a x_t + \kappa \sum_{j=1}^K q_j y_{j,t} + w_{t+1} \right). \tag{10}$$

Stacking the pricing error x_t and the K internal cohort states into the state vector $\mathbf{z}_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top \in \mathbb{R}^{K+1}$, the joint economy forms an autonomous discrete-time linear dynamical system:

$$\mathbf{z}_{t+1} = J\mathbf{z}_t + \mathbf{b} w_{t+1}, \tag{11}$$

where the closed-loop system Jacobian matrix $J \in \mathbb{R}^{(K+1)\times(K+1)}$ is given by:

$$J = \begin{bmatrix} a & \kappa q_1 & \kappa q_2 & \cdots & \kappa q_K \\ \eta_1 a & 1-\eta_1+\eta_1\kappa q_1 & \eta_1\kappa q_2 & \cdots & \eta_1\kappa q_K \\ \eta_2 a & \eta_2\kappa q_1 & 1-\eta_2+\eta_2\kappa q_2 & \cdots & \eta_2\kappa q_K \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \eta_K a & \eta_K\kappa q_1 & \eta_K\kappa q_2 & \cdots & 1-\eta_K+\eta_K\kappa q_K \end{bmatrix}, \tag{12}$$

and $\mathbf{b} = (1, \eta_1, \eta_2, \ldots, \eta_K)^\top$.

Single-Cohort Stability Boundary. In the single-cohort case ($K = 1$), defining total loop feedback gain $Q \equiv \kappa q_1$, the characteristic polynomial of J is:

$$P(\lambda) = \det(\lambda I - J) = \lambda^2 - (a + 1 - \eta + \eta Q)\,\lambda + (1-\eta)a. \tag{13}$$

By the Jury stability criterion, all roots of $P(\lambda)$ lie strictly inside the open unit disk $\{|\lambda| < 1\}$ if and only if $|P(0)| < 1$, $P(1) > 0$, and $P(-1) > 0$. Here $|P(0)| = (1-\eta)a < 1$ holds for all $a \in [0, 1)$ and $\eta \in (0, 1)$. Evaluating at $\lambda = 1$ and $\lambda = -1$:

$$P(1) = (1 - a - Q)\eta, \qquad P(-1) = (2-\eta)(1+a) + \eta Q. \tag{14}$$

For the positive-feedback economy ($Q \geq 0$), $P(-1) > 0$ is strictly satisfied, so Schur stability is strictly equivalent to:

$$S \equiv a + Q < 1, \qquad \text{or equivalently} \quad \mathcal{L} \equiv \frac{Q}{1-a} < 1, \tag{15}$$

where S is total closed-loop persistence and $\mathcal{L}$ is the loop ratio. In the general signed-feedback economy ($Q \in \mathbb{R}$), the full Jury stability interval is $-\frac{(2-\eta)(1+a)}{\eta} < Q < 1 - a$, where the lower bound prevents period-doubling oscillatory explosion. For positive feedback the same $+1$ boundary also holds with a lagged update; signed-feedback Jury conditions must be recomputed: if execution features a discrete one-period lag ($x_{t+1} = ax_t + y_t + w_{t+1}, y_{t+1} = (1-\eta)y_t + \eta Q x_t$), evaluating $P_{\text{lag}}(1) = (1 - a - Q)\eta > 0 \iff S \equiv a + Q < 1$, identically.

2.2 Theorem 1: Merged Rate-Weight Equivalence and $\Omega(\delta^{-4})$ Separation

A central question in algorithmic market surveillance is whether an econometrician observing public trade executions and prices can identify the individual algorithmic strategies active

in the market.

Theorem 2.3 (Merged Rate-Weight Equivalence and Minimax Separation Complexity)**.** *Let price impact $\kappa > 0$ and anchor $a \in [0, 1)$ be known. Compare zero-initial-state responses with process innovations set to zero and adaptation rates strictly in $(0, 1)$. The merged support excludes rates whose total weight cancels to zero.*

1. ***Exact Equivalence Class:*** *Two market economies $\mathcal{E}_A = \{(q_i^A, \eta_i^A)\}_{i=1}^{K_A}$ and $\mathcal{E}_B = \{(q_i^B, \eta_i^B)\}_{i=1}^{K_B}$ generate mathematically identical public price responses x_t for all $t \geq 0$ under all public probing sequences $\{u_t^{probe}\}_{t\geq 0}$ if and only if their merged rate–weight measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A = \sum_{j:\eta_j^B=\eta} q_j^B \qquad \forall \eta \in (0, 1). \tag{16}$$

   The minimal state-space realization dimension is $1 + |\operatorname{supp}(\mathcal{M})|$, where the support counts distinct rates with nonzero merged weights; this statement excludes instantaneous rates. With these nonzero weights there are no cancellations at the distinct cohort poles.

2. ***Minimax Separation Complexity:*** *Consider testing whether a noiseless-state market ($w_t = 0$) consists of a single merged cohort with rate $\bar{\eta}$ versus two split cohorts with rates $\bar{\eta} \pm \delta$ and identical fixed nonzero total feedback Q, with $\sigma_v > 0$ and $0 < \delta_0 < \min(\bar{\eta}, 1 - \bar{\eta})$. Assume that the closed-loop family lies in a compact parameter set with $\rho(J) \leq 1 - \varepsilon$ uniformly for $|\delta| \leq \delta_0$, with $\varepsilon > 0$. Under deterministic probing sequences satisfying power constraint $\frac{1}{T}\sum_{t=0}^{T-1} u_t^2 \leq \mathcal{P}_{probe}$ and independent additive Gaussian observation noise $\tilde{x}_t = x_t + v_t$ with $v_t \sim \mathcal{N}(0, \sigma_v^2)$, any deterministic/open-loop probing sequence satisfying the stated average-power constraint distinguishing the two economies with worst-case probability of error at most $\alpha \in (0, 1/4)$ requires an observation budget T satisfying:*

$$T \geq \frac{C(\alpha, \sigma_v, \mathcal{P}_{probe}, \mathcal{S}_\varepsilon)}{\delta^4} = \Omega(\delta^{-4}), \tag{17}$$

   where $C(\alpha, \sigma_v, \mathcal{P}_{probe}, \mathcal{S}_\varepsilon) > 0$ is a constant independent of δ.

Proof. **Part 1 (Equivalence Class):** Taking z-transforms of the pricing equation (4) and cohort updates (9), we have $X(z) = \frac{\kappa}{z-a}U(z)$ and $Y_i(z) = \frac{\eta_i z}{z-(1-\eta_i)}X(z)$. Summing across cohorts, total endogenous demand is $U_{\text{end}}(z) = \sum_{i=1}^K q_i Y_i(z) = \left(\sum_{i=1}^K \frac{q_i\eta_i z}{z-(1-\eta_i)}\right) X(z)$. Applying feedback closure $X(z) = \frac{\kappa}{z-a}(U_{\text{end}}(z) + U_{\text{probe}}(z))$ yields the public transfer function from

probe to price:

$$H(z) \equiv \frac{X(z)}{U_{\text{probe}}(z)} = \frac{\kappa}{z - a - \kappa G(z)}, \qquad G(z) \equiv z \sum_{r \in \mathcal{R}} \frac{c_r}{z - r}, \tag{18}$$

where the set of poles is $\mathcal{R} = \{1 - \eta : \eta \in \text{supp}(\mathcal{M})\}$, with merged coefficients $c_r = \eta \sum_{i:\eta_i=\eta} q_i$. Because $\frac{zc_r}{z-r} = c_r + \frac{rc_r}{z-r}$, the pole is $r = 1 - \eta$, and its residue is rc_r. Since $r \neq 0$ for $\eta \in (0, 1)$, equality of poles and residues is equivalent to equality of the merged coefficients c_r. By the fundamental theorem of algebra and the uniqueness of partial fraction expansions of proper rational functions on $\mathbb{C}$, equality of H_A and H_B implies equality of G_A and G_B by $G = (z - a - \kappa/H)/\kappa$. Uniqueness of the poles and residues of G therefore determines its nonzero merged weights. Thus, any two economies with identical merged rate-weights c_r produce identical transfer functions, and hence identical price impulse responses. For a rational single-input/output transfer function, the minimal realization degree is the denominator degree after cancellation (Gilbert, 1963), which is $1 + |\,\text{supp}(\mathcal{M})|$ for dynamic cohorts $\eta_i \in (0, 1)$.

Part 2 (Minimax Sample Complexity): Let $\mathcal{E}_0$ have single cohort $(\bar{q}, \bar{\eta})$, and $\mathcal{E}_1$ have two cohorts with rates $\bar{\eta}_1 = \bar{\eta} - \delta$ and $\bar{\eta}_2 = \bar{\eta} + \delta$ with weights $q_1 = q_2 = \bar{q}/2$, so total feedback $Q = \kappa\bar{q}$ is preserved. The endogenous demand transfer per unit price is $G_0(z) = \frac{\bar{q}\bar{\eta}z}{z-(1-\bar{\eta})}$ under $\mathcal{E}_0$, and $G_\delta(z) = \frac{\bar{q}z}{2}[\frac{\bar{\eta}-\delta}{z-(1-\bar{\eta}+\delta)} + \frac{\bar{\eta}+\delta}{z-(1-\bar{\eta}-\delta)}]$ under $\mathcal{E}_1$. Due to bilateral symmetry $\bar{\eta} \pm \delta$, the perturbation $\Delta G(z) \equiv G_\delta(z) - G_0(z)$ is strictly even in δ, so all odd-order terms in δ vanish identically: $\Delta G(z) = G_2(z)\delta^2 + \mathcal{O}(\delta^4)$. Writing $D(z) \equiv z - a - \kappa G(z)$, the closed-loop price transfer discrepancy is:

$$H_\delta(z) - H_0(z) = \frac{\kappa}{D_\delta(z)} - \frac{\kappa}{D_0(z)} = \frac{\kappa^2 \Delta G(z)}{D_0(z) D_\delta(z)} = H_0(z) H_\delta(z) \Delta G(z). \tag{19}$$

Assume the parameter family belongs to the stated compact parameter family $\mathcal{S}_\varepsilon$ satisfying $\rho(J) \leq 1 - \varepsilon$ for some fixed $\varepsilon > 0$ for all $|\delta| \leq \delta_0$. Under this condition, all closed-loop poles lie strictly inside the disk $\{|z| \leq 1 - \varepsilon\}$. Thus $H_\delta(z) - H_0(z)$ is analytic on $\{|z| \geq 1\}$. Choose $R \in (1 - \varepsilon, 1)$. Symmetry of the alternatives and compactness bound $H_\delta - H_0$ by $C_R\delta^2$ uniformly on $|z| = R$. Cauchy's coefficient formula then bounds its impulse coefficients by $C_R\delta^2 R^t$, with the constant absorbing the time-index convention. Summing the sequence yields bounded ℓ_1 norm $\|\Delta h\|_{\ell_1} = \sum_{t=1}^{\infty} |\Delta h_t| \leq C_H \cdot \delta^2$ with constant $C_H < \infty$ strictly independent of δ. Under probe sequence $\{u_t\}_{t=0}^{T-1}$ with power budget $\frac{1}{T}\sum_{t=0}^{T-1} u_t^2 \leq \mathcal{P}_{\text{probe}}$ (so $\|u\|_{\ell_2} \leq \sqrt{T\mathcal{P}_{\text{probe}}}$), the deterministic output price discrepancy is $\Delta x = \Delta h * u$. By Young's convolution inequality, $\|\Delta x\|_{\ell_2} \leq \|\Delta h\|_{\ell_1}\|u\|_{\ell_2} \leq C_H\delta^2\sqrt{T\mathcal{P}_{\text{probe}}}$, yielding $\sum_{t=1}^{T}(\Delta x_t)^2 \leq C_H^2 \mathcal{P}_{\text{probe}} T\delta^4$. Under controlled probing with independent additive Gaussian observation

noise $v_t \sim \mathcal{N}(0, \sigma_v^2)$ on prices $\tilde{x}_t = x_t + v_t$, the noise covariance $\sigma_v^2 I_T$ is identical across both hypotheses, and the trajectory mean discrepancy is deterministic Δx. The exact Kullback–Leibler (KL) divergence is therefore:

$$D_{\mathrm{KL}}\left(P_0^T \,\|\, P_1^T\right) = \frac{1}{2\sigma_v^2}\sum_{t=1}^{T}(\Delta x_t)^2 \le \frac{C_H^2 \mathcal{P}_{\mathrm{probe}}}{2\sigma_v^2} T\delta^4 \equiv T \cdot C_0 \cdot \delta^4. \tag{20}$$

By Le Cam's two-point inequality (Le Cam, 1986), the minimax probability of error across the two hypotheses satisfies:

$$\inf_{\Psi}\max_{i\in\{0,1\}} \mathbb{P}_i(\Psi \neq i) \ge \frac{1}{2}\left(1 - \|P_0^T - P_1^T\|_{\mathrm{TV}}\right) \ge \frac{1}{2}\left(1 - \sqrt{\frac{1}{2}D_{\mathrm{KL}}(P_0^T \,\|\, P_1^T)}\right). \tag{21}$$

To achieve worst-case error bounded by $\alpha \in (0, 1/4)$, we must have $\|P_0^T - P_1^T\|_{\mathrm{TV}} \ge 1 - 2\alpha > 1/2$, which requires $D_{\mathrm{KL}}(P_0^T \,\|\, P_1^T) \ge 2(1-2\alpha)^2$. Substituting $D_{\mathrm{KL}} \le T \cdot C_0 \cdot \delta^4$ yields:

$$T \ge \frac{2(1-2\alpha)^2}{C_0\,\delta^4} \equiv \frac{C(\alpha, \sigma_v, \mathcal{P}_{\mathrm{probe}}, \mathcal{S}_\varepsilon)}{\delta^4} = \Omega(\delta^{-4}), \tag{22}$$

completing the proof. □

Remark 2.4 (Scope of the Separation Bound). Merged rates identify aggregate weights, not the count or identities of participants sharing a rate in the reset linear experiment. The symmetric split construction makes the leading response difference quadratic in δ, giving a fourth-order testing lower bound with fixed stability margin and noise/power assumptions. This does not identify dollar allocations, supply an upper bound, or justify a literal ratio of detection horizons between $\delta = 0.05$ and a unit perturbation outside the local family.

2.3 Theorem 2: A Finite-Window Price-Surveillance Bound

The following experiment isolates a statistical boundary under passive price observations. Its parameter s is the scalar transition coefficient, distinct from the algebraic index $S = a + Q$ of a general multivariate market. Applying the result to a hidden multivariate market requires a valid observation-law reduction; no such reduction is inferred solely from a common unit-root boundary.

Theorem 2.5 (Finite-Window Abstention in a Scalar Gaussian Experiment)**.** *Observe* $\mathcal{D}_T = (x_1, \ldots, x_T)$ *where* $x_0 = 0$ *and* $x_t = sx_{t-1} + w_t$, $w_t \sim \mathcal{N}(0, \sigma^2)$ *independently,* $\sigma > 0$, *and*

$s \in (0,2)$. *Let a possibly randomized rule* Ψ *take values Stable, Unstable, or Abstain, with*

$$\sup_{s<1} P_s(\Psi = \text{Unstable}) \le \alpha, \qquad \sup_{s\ge 1} P_s(\Psi = \text{Stable}) \le \alpha, \quad 0 < \alpha < 1/2.$$

Choose $0 < c_0 < 1$ *with* $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. *For every integer* $T \ge 1$, *on* $\mathcal{B}_T = (1 - c_0/T, 1 + c_0/T)$,

$$\inf_{s\in\mathcal{B}_T} P_s(\Psi = \text{Abstain}) \ge 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0. \tag{23}$$

For every finite T, *the minimax worst-case error of binary rules across* $s \in (0,2)$ *is at least* $1/2$.

Proof. Pair any $s_0 \in (1 - c_0/T, 1)$ with $s_1 = s_0 + c_0/T \in (1, 1 + c_0/T)$; reverse the pairing for a point on the other side, including $s_1 = 1$. Gaussian transition densities give

$$D_{\mathrm{KL}}(P_{s_0} \| P_{s_1}) = \frac{(s_1 - s_0)^2}{2\sigma^2} \sum_{t=1}^{T} E_{s_0} x_{t-1}^2.$$

For either paired coefficient, $|s| \le 1 + c_0/T$ and $E_s x_{t-1}^2 = \sigma^2 \sum_{j=0}^{t-2} s^{2j} \le \sigma^2 (t-1) e^{2c_0}$. Thus the KL divergence in either direction is at most $c_0^2 e^{2c_0}/4$ and Pinsker's inequality bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Unstable}) \le \alpha$ and $P_{s_0}(\text{Stable}) \le P_{s_1}(\text{Stable}) + \text{TV} \le \alpha + \text{TV}$, yielding the stated abstention bound. The reverse pairing is identical.

For binary rules, the Gaussian path densities are continuous in s in total variation at each finite T. Let $s^- \uparrow 1$ and $s^+ \downarrow 1$. Le Cam's two-point bound gives worst-case error at least $(1 - \|P_{s^-} - P_{s^+}\|_{\mathrm{TV}})/2 \to 1/2$. This is a worst-case statement over arbitrarily close alternatives; it is not a 50% error claim at every fixed separated parameter. □

The theorem forces a positive probability of withholding a classification under the specified uniform error guarantee. It does not require abstention on every sample path. Initial conditions, extra measurements, and observation noise can change the inference problem. In particular, a fully observed noiseless cohort recursion need not admit the common Lebesgue densities assumed by an informal reduction to a scalar price likelihood. Proposition G.1 gives a constructive two-step identification result under an exact single-cohort state/demand contract, resolving that case explicitly.

2.4 Theorem 3: Intermediary Solvency and the Kreiss Constant Bound

We now confront the fundamental divergence between price stability and financial intermediary solvency. Spectral radius describes asymptotic decay in the linear approximation. Loss exposure additionally depends on the shock set, output operator and finite-time magnification.

The Non-Normal Market Jacobian. Consider the closed-loop Jacobian matrix J from (12). In financial markets, price impact and memory adaptation are asymmetric: order flow directly affects prices through contemporaneous impact κ, while price innovations affect memory through learning rate η. As a result, $JJ^\top - J^\top J \neq 0$: the closed-loop market Jacobian J is generically **non-normal** under asymmetric price-impact and learning parameters (except on a measure-zero algebraic subvariety). While normal matrices have orthogonal eigenvectors that diagonalize under unitary transformations, non-normal matrices have non-orthogonal (nearly parallel) eigenvectors. Consequently, even when all eigenvalues are strictly inside the unit circle ($|\lambda_i(J)| \leq \rho(J) < 1$), matrix powers J^t can undergo massive transient amplification before decaying asymptotically as $t \to \infty$ (Trefethen and Embree, 2005).

Leveraged Dealer Balance-Sheet Model. Let E_0 be unencumbered equity and $0 < \gamma \leq 1$ a chosen allocation to a modeled absolute loss/exposure budget. Define

$$B = \gamma E_0. \tag{24}$$

This budget is not leverage-created capital. A separate margin or leverage allowance requires its own balance-sheet rule. Fix state units and an output operator C mapping state disturbance to declared dollar exposures. A single row measures one loss channel; multiple commensurately scaled rows define a vector exposure budget. The homogeneous initial-shock calculation below omits ongoing innovations and describes a stylized budget breach, not a complete stochastic solvency model.

Let $z_t \in \mathbb{R}^n$ denote the state vector of the closed-loop economy evolving as $z_t = J^t z_0$, and let $C \in \mathbb{R}^{m \times n}$ denote the balance-sheet observation operator that maps the state z_t into the declared dollar exposures (one output for a scalar dealer loss):

$$\text{Loss}_t = \|Cz_t\|_2 = \|CJ^t z_0\|_2. \tag{25}$$

A modeled loss-budget breach occurs if $\sup_{t \geq 0} \|CJ^t z_0\|_2 > B$.

Theorem 2.6 (Dealer Solvency and the Kreiss Constant Bound)**.** *Let $z_t = J^t z_0$ with $\rho(J) < 1$, balance-sheet buffer $B > 0$, and observation matrix C.*

1. ***Exact Worst-Case Safe Radius:*** *The exact maximum shock radius r^*_{exact} guaranteeing that no initial disturbance with $\|z_0\|_2 \le r^*_{exact}$ breaches the declared output budget in any direction is:*

$$r^*_{exact} = \frac{B}{\sup_{t\ge 0} \|CJ^t\|_2} \equiv \frac{B}{M_C}, \tag{26}$$

where $M_C \equiv \sup_{t\ge 0} \|CJ^t\|_2$ is peak output magnification. Assume $C \neq 0$. For $C = I$, the budget constrains the Euclidean norm of the output vector rather than a scalar signed loss.

2. ***Kreiss Lower and Upper Bounds:*** *Under full state observation ($C = I$), the Kreiss constant of matrix J, defined on the resolvent outside the unit disk:*

$$\mathcal{K}(J) \equiv \sup_{|z|>1} (|z| - 1)\|(zI - J)^{-1}\|_2, \tag{27}$$

satisfies $\mathcal{K}(J) \le \sup_{t\ge 0} \|J^t\|_2 \le e \cdot n \cdot \mathcal{K}(J)$ for any $J \in \mathbb{R}^{n\times n}$. Consequently:

$$r_{conservative} \equiv \frac{B}{e \cdot n \cdot \mathcal{K}(J)} \le r^*_{exact} \le \frac{B}{\mathcal{K}(J)}. \tag{28}$$

3. ***Complete Failure of Spectral Radius:*** *For any stable spectral radius $\rho \in (0, 1)$ and any arbitrarily small buffer $B > 0$, there exists a parameter family of non-normal matrices J_θ and an observation operator $C = [1, 0]$ exposed to the amplified coordinate such that $\rho(J_\theta) = \rho$ for all θ, while the exact safe radius collapses to zero:*

$$\lim_{\theta\to\infty} r^*_{exact}(J_\theta) = 0. \tag{29}$$

Proof. **Part 1:** By the Cauchy–Schwarz inequality for operator norms, the maximum loss across all initial shock vectors with $\|z_0\|_2 \le r$ is:

$$\sup_{\|z_0\|_2\le r} \sup_{t\ge 0} \|CJ^t z_0\|_2 = r \cdot \sup_{t\ge 0} \|CJ^t\|_2 = r \cdot M_C. \tag{30}$$

Requiring that the worst-case loss does not breach capital buffer B imposes $r \cdot M_C \le B$, which holds if and only if $r \le B/M_C \equiv r^*_{\text{exact}}$.

Part 2: The lower bound $\mathcal{K}(J) \le \sup_{t\ge 0} \|J^t\|_2$ follows directly from the Neumann series

expansion of the resolvent: for $|z| > 1$, $(zI - J)^{-1} = \sum_{t=0}^{\infty} z^{-(t+1)} J^t$. Taking norms:

$$\|(zI - J)^{-1}\|_2 \leq \sum_{t=0}^{\infty} |z|^{-(t+1)} \|J^t\|_2 \leq \sup_{t \geq 0} \|J^t\|_2 \sum_{t=0}^{\infty} |z|^{-(t+1)} = \frac{\sup_{t \geq 0} \|J^t\|_2}{|z| - 1}. \tag{31}$$

Multiplying both sides by $(|z| - 1)$ and taking the supremum over $|z| > 1$ yields $\mathcal{K}(J) \leq \sup_{t \geq 0} \|J^t\|_2$. The upper bound $\sup_{t \geq 0} \|J^t\|_2 \leq e \cdot n \cdot \mathcal{K}(J)$ is Spijker's Theorem (Spijker, 1991), which establishes the optimal dimensional constant $e \cdot n$ for the discrete Kreiss matrix theorem. Inverting the inequality chain $\mathcal{K}(J) \leq M_I \leq e \cdot n \cdot \mathcal{K}(J)$ and multiplying by B immediately yields (28).

Part 3: Consider the 2×2 family of matrices parameterized by $\theta > 0$:

$$J_\theta = \begin{bmatrix} \rho & \theta \\ 0 & \rho \end{bmatrix}, \qquad \rho \in (0, 1). \tag{32}$$

The eigenvalues are $\lambda_1 = \lambda_2 = \rho$, so $\rho(J_\theta) = \rho < 1$ for all θ. Computing the t-th power:

$$J_\theta^t = \begin{bmatrix} \rho^t & t\theta\rho^{t-1} \\ 0 & \rho^t \end{bmatrix}. \tag{33}$$

Setting $C = [1, 0]$ and initial shock $z_0 = [0, 1]^\top$, the loss at time t is $CJ_\theta^t z_0 = t\theta\rho^{t-1}$. Already at $t = 1$, the unit shock $z_0 = [0, 1]^\top$ has loss θ, so $M_C \geq \theta$ and $r^*_{\text{exact}} \leq B/\theta$. No continuous-time approximation to the discrete peak is required. Taking $\theta \to \infty$, $r^*_{\text{exact}} \to 0$ while the spectral radius remains frozen at ρ, completing the proof. □

Remark 2.7 (Interpretation and Coordinates). Part 3 concerns the unrestricted class of stable linear matrices. It does not assert that every Jordan-family coupling has an admissible structural market parameterization. Non-normality alone does not imply a large financially relevant transient: the loss operator and disturbance directions must expose it. Rescaling states changes Euclidean shock balls and transient norms unless the metric is transformed consistently. Fixed physical units or a justified weighted norm are therefore necessary for financial comparisons.

Under $C = I$, $B/(en\mathcal{K}(J))$ is a guaranteed conservative lower bound on the allowable radius; $B/\mathcal{K}(J)$ is an upper bound, not a safe gating threshold. For a general output matrix, $M_C \leq \|C\|_2 en\mathcal{K}(J)$ yields a conservative radius $B/(\|C\|_2 en\mathcal{K}(J))$.

2.5 Pseudospectral Interpretation

For $\epsilon > 0$, define $\sigma_\epsilon(J) = \{z : \|(zI - J)^{-1}\|_2 \geq \epsilon^{-1}\}$, including the spectrum by continuity, and $\alpha_\epsilon(J) = \sup_{z \in \sigma_\epsilon(J)} |z|$. The resolvent form of the Kreiss constant is equivalently

$$\mathcal{K}(J) = \sup_{\epsilon > 0} \frac{\alpha_\epsilon(J) - 1}{\epsilon}.$$

This relates transient growth to pseudospectral protrusion outside the unit disk in a specified norm (Trefethen and Embree, 2005). It is a geometric interpretation of established operator theory, not a measured market contour. We report actual matrix-power output magnification in Section 5, with explicit matrices and loss coordinates.

2.6 Multi-Asset Generalization and Cross-Impact Kreiss Inflation

Modern market makers operate simultaneously across multiple correlated instruments (e.g., BTC, ETH, and altcoins; or equity index components and ETFs). In multi-asset environments, cross-asset price impact creates additional non-normal coupling.

Let $\mathbf{x}_t \in \mathbb{R}^d$ denote the vector of pricing errors across d traded assets, and let $\mathbf{u}_t \in \mathbb{R}^d$ denote aggregate order flows. The multi-asset price formation equation is:

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{K}\mathbf{u}_t + \mathbf{w}_{t+1}, \tag{34}$$

where $\mathbf{A} = \mathrm{diag}(a_1, \ldots, a_d) \in \mathbb{R}^{d \times d}$ collects direct price persistences, and $\mathbf{K} \in \mathbb{R}^{d \times d}$ is the cross-asset price impact matrix. In general, $\mathbf{K}$ is asymmetric ($\mathbf{K} \neq \mathbf{K}^\top$): institutional order flow in a benchmark asset (such as BTC or SPY) exerts cross-impact on satellite assets, but the reverse flow produces much smaller reactions.

Each asset is traded by adaptive momentum flow with learning rates $\mathbf{H} = \mathrm{diag}(\eta_1, \ldots, \eta_d)$ and feedback gain $\mathbf{Q} \in \mathbb{R}^{d \times d}$:

$$\mathbf{y}_{t+1} = (\mathbf{I} - \mathbf{H})\mathbf{y}_t + \mathbf{H}\mathbf{x}_{t+1}, \qquad \mathbf{u}_t = \mathbf{Q}\mathbf{y}_t. \tag{35}$$

Stacking $\mathbf{z}_t = (\mathbf{x}_t^\top, \mathbf{y}_t^\top)^\top \in \mathbb{R}^{2d}$, the multi-asset closed-loop Jacobian is:

$$J_{\text{multi}} = \begin{bmatrix} \mathbf{A} & \mathbf{K}\mathbf{Q} \\ \mathbf{H}\mathbf{A} & \mathbf{I} - \mathbf{H} + \mathbf{H}\mathbf{K}\mathbf{Q} \end{bmatrix} \in \mathbb{R}^{2d \times 2d}. \tag{36}$$

Proposition 2.8 (Cross-Impact Kreiss Inflation and Safe Radius Bound in Directed Networks)**.** *Consider a multi-asset economy with directed block-triangular leader-satellite cross-*

impact ($K_{12} = 0, K_{21} > 0$), diagonal feedback matrices $\mathbf{Q} \geq 0$, anchor matrices $\mathbf{A} \geq 0$, adaptation rates $\mathbf{H} \in (0,1)^d$, nonnegative cross-impact $\mathbf{K} \geq 0$, and a nonzero portfolio output operator $C \geq 0$. Assume $\rho(J_{multi}) < 1$. Let J_{diag} denote the decoupled Jacobian where $K_{21} = 0$. The directed coupling introduces off-diagonal block feedforward, inflating the Kreiss constant: $\mathcal{K}(J_{multi}) \geq \mathcal{K}(J_{diag})$. Consequently, the Kreiss-based conservative safe radius bound contracts: $r_{conservative}(J_{multi}) \equiv \frac{B}{\|C\|_2 e(2d)\mathcal{K}(J_{multi})} \leq r_{conservative}(J_{diag})$. For the exact safe shock radius $r^*_{exact}(J) \equiv B/M_C(J)$, because all constituent blocks of J are entrywise non-negative in the momentum economy, matrix powers J^t_{multi} are entrywise non-decreasing in K_{21}, yielding $M_C(J_{multi}) \geq M_C(J_{diag})$ and exact radius contraction $r^*_{exact}(J_{multi}) \leq r^*_{exact}(J_{diag})$. Coupled powers contain sums of leader-to-satellite propagation terms; their output magnitude depends on the fixed loss coordinates. No market-calibrated contraction percentage is asserted.

Proof. After ordering states by asset, the coupled matrix is block triangular with the same diagonal blocks as the decoupled direct sum. The diagonal compressions of its resolvent are unchanged. Its operator norm is therefore at least the maximum norm of those diagonal blocks, which is the decoupled resolvent norm; taking the Kreiss supremum gives the first inequality. Nonnegative powers and a nonnegative output operator are entrywise nondecreasing in the directed coupling. The spectral norm is monotone for nonnegative matrices, giving $M_C(J_{\text{multi}}) \geq M_C(J_{\text{diag}})$. The budget-radius inequalities follow with the same B and C. □

Proposition 2.8 demonstrates that multi-asset diversification does not automatically insulate liquidity providers from feedback fragility. On the contrary, asymmetric cross-impact networks (such as Bitcoin leading altcoins or large-cap equities driving ETFs) introduce feedforward coupling that changes finite-time propagation in matrix powers J^t, inflating peak transient loss M_C and potentially reducing the radius under the stated assumptions. This is a conditional network result rather than a calibrated multi-asset estimate.

3 Data, Measurements, and the Observable Estimand

3.1 Observed Price Proxies and Flow

The theoretical state $x_t = p_t - p^*_t$ is not observed in these datasets. For the primary Coinbase analysis we use $p_b = 10{,}000 \log(P_b)$, where P_b is the recorded bid–ask midpoint at the end of a 10-second bin, and an operational reference $m_b = (1-\alpha)m_{b-1} + \alpha p_b$, $\alpha = 2/(300/10+1)$. The empirical state is $x^{\text{EMA}}_b = p_b - m_b$. It is a deviation from a trailing filter, not a measured fundamental pricing error. A separate robustness state is the raw return $r_b = p_b - p_{b-1}$.

Signed executed flow f_b is the recorded taker-buy minus taker-sell quote-currency notional aggregated in bin b. Repeated renders of the same book/trade record receive zero new flow following the recorded correction protocol. Price and flow both refer to information observed by the bin's end. Existing source timestamps and rolling trade counters only provide the recorder's timing resolution; they do not establish exchange-level execution latency.

3.2 A Reduced-Form Diagnostic

For an observable state $v_b \in \{x_b^{\mathrm{EMA}}, r_b\}$ and observed flow f_b, fit

$$\begin{bmatrix} v_{b+1} \\ f_{b+1} \end{bmatrix} = d + J \begin{bmatrix} v_b \\ f_b \end{bmatrix} + e_{b+1}, \qquad Q = \frac{J_{12}J_{21}}{1 - J_{22}}, \quad S = J_{11} + Q. \tag{37}$$

Both regressors are observed at time b. The legacy implementation instead pairs v_b with f_{b+1}, creating an overlapping forward-flow statistic. That alignment is retained only as a sensitivity comparison, not as a real-time observable state.

Order-flow or price-unit rescaling conjugates J by a diagonal scaling matrix; the product $J_{12}J_{21}$, Q, S, and eigenvalues are unchanged. However, scale invariance does not establish causality, invariance to filtering, or equivalence to a latent economic state. Non-normal Euclidean norms additionally require fixed coordinates or a specified weighted norm.

For $J_{11}, J_{22} \in [0, 1)$ and $J_{12}J_{21} \geq 0$, $S < 1$ characterizes the positive real-root boundary and Schur stability of this two-state fit. Outside those sign restrictions, $S < 1$ alone is insufficient; we always compute $\rho(J)$ directly. A multivariate fit to raw returns is not interchangeable with a fit to a filtered price level.

3.3 Filters, Missingness, and Sessions

For a martingale price with no price-to-flow feedback, the EMA deviation satisfies $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(x_b^{\mathrm{EMA}} + \Delta p_{b+1})$. Thus an anchor near $1-\alpha = 0.93548$ can arise mechanically. Positive feedback must be interpreted against that construction and explicit no-adaptation controls.

The revision excludes transitions crossing UTC dates, timestamp gaps, or equity session boundaries. It does not backward-fill missing prices or manufacture unobserved trading intervals. The primary crypto segments receive five EMA spans of initialization burn-in and require at least 500 retained observations. The equity analysis uses a separate rule: at least 600 raw session rows and a 500-event burn-in, so some retained equity fits are smaller than 500 observations. Segment estimates on the same date are combined using transition counts before date-level averaging and Student-t intervals. The 13 Coinbase calendar dates include

partial first and last dates. These intervals summarize variation across observed dates; they do not provide a long-horizon or independent external replication guarantee.

3.4 Actual Dataset Scope

Coinbase BTC, ETH, and ADA snapshots cover April 7–19, 2021. Binance's 768 saved asset-day fits cover twice-monthly dates in 2021–2024, but their generator reads 1-second trade-price klines and estimates at 10-second spacing with a 300-second EMA and forward flow. Klines do not contain bid–ask midpoints, queue identities, or full L3 books. The 66.3 million nominal seconds in the previous description are not the number of fitted transitions: the saved panel contains 6,519,552 fitted state rows, with one fewer transition per asset-day. Its original raw archives are not present in the inspected cache.

FI-2010 is an event-based book benchmark (Ntakaris et al., 2018). The local representation contains 272,941 rows with supplied STOCK and DAY fields. Those fields define this reanalysis; the original identifier mapping has not been independently authenticated. The January BTC depth file covers January 9–20, 2023 rather than the entire month. It contains quote depth, and the revision uses quote-update OFI, not authenticated executed trades. Its provider attribution is not independently established here. It was already examined before this revision, so it is a secondary sample rather than an untouched prospective holdout.

4 Observable Evidence and Fee Sensitivity

4.1 Fresh Coinbase Reanalysis

The following table is generated from fresh end-bin-aligned fits of the existing corrected recorder data. The primary filtered-state estimates are small and positive on average. Their magnitude changes substantially with alignment; raw-return intervals include zero for all three assets.

The comparison cautions against interpreting the larger legacy estimates as the gain of a specific learning rule. Anonymous data cannot distinguish algorithmic response from other persistent or informational channels. The primary EMA-state intervals describe aggregate observed relations; they do not supply a structural trader attribution.

4.2 No-Adaptation Controls

We generate 60 independent 6,000-second paths for each declared control, aggregate at 10 seconds, and apply the same 300-second EMA and burn-in. The four controls are martingale

Table 1: Timing and state sensitivity on the recorded Coinbase sample. Feedback intervals use date-level Student-t variation; maximum radius is over fitted day segments.

Asset	State / flow	Dates	$\hat{Q}$ [95% interval]	Mean $\hat{S}$	Max ρ
BTC	EMA / observed	13	0.00194 [0.00032, 0.00355]	0.9426	0.9587
BTC	EMA / forward	13	0.01252 [0.00653, 0.01851]	0.9419	0.9586
BTC	Return / observed	13	0.00121 [-0.00067, 0.00308]	0.0569	0.3092
ETH	EMA / observed	13	0.00098 [0.00006, 0.00190]	0.9392	0.9574
ETH	EMA / forward	13	0.01130 [0.00806, 0.01454]	0.9383	0.9586
ETH	Return / observed	13	0.00033 [-0.00035, 0.00101]	0.0274	0.3597
ADA	EMA / observed	13	0.00062 [0.00016, 0.00108]	0.9350	0.9515
ADA	EMA / forward	13	0.00721 [0.00564, 0.00877]	0.9341	0.9521
ADA	Return / observed	13	0.00000 [-0.00094, 0.00094]	0.0144	0.2974

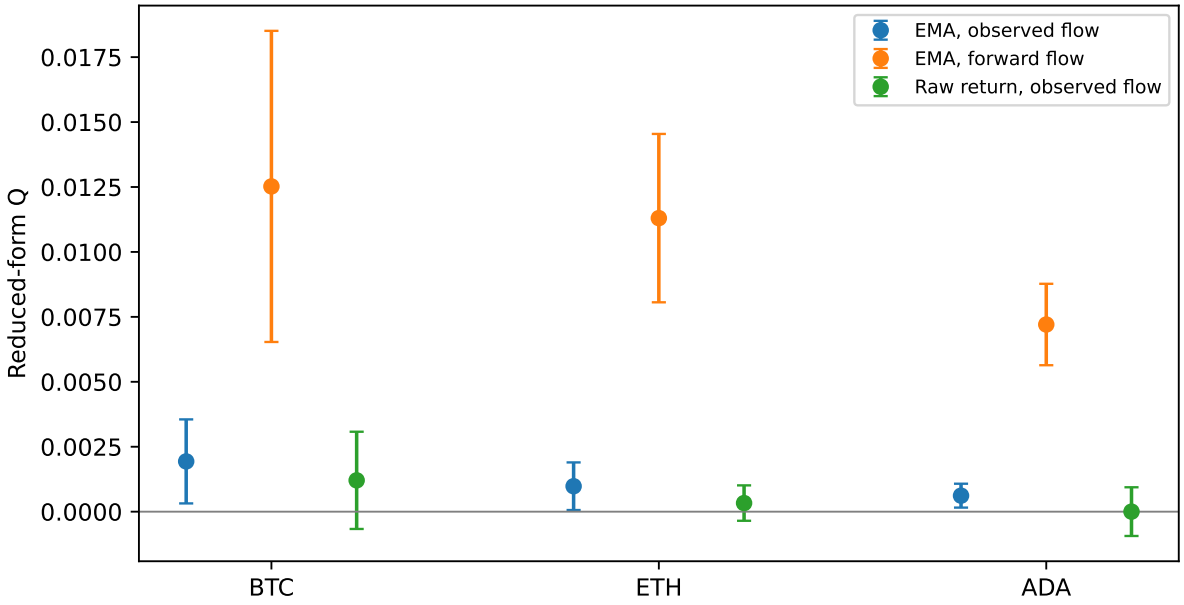


Figure 1: Feedback diagnostics change with observable state and timing. All plotted intervals are regenerated from the revision's daily estimates.

prices with independent flow; persistent exogenous flow independent of prices; persistent exogenous flow with price impact; and common price–flow news. None has a price-to-flow adaptive rule. These are bounded diagnostic experiments, not calibrated real-market null distributions or evidence that all possible confounding has been excluded.

The controls show that filter construction, persistence, price impact, and information timing must be specified jointly. A diagnostic's sign is not an identification theorem. The historical estimator-validation archive also contains a no-feedback persistent-flow example with positive VAR feedback; that unfavorable result is retained in the replication materials.

Table 2: No-adaptation controls. Standard deviations describe Monte Carlo variation across 60 independent paths, not standard errors for the market estimates.

Control	Observed mean Q	SD	Forward mean Q	SD
Independent martingale	-0.00018	0.00076	-0.00006	0.00086
Persistent exogenous flow	0.00027	0.00213	0.00024	0.00209
Order splitting with impact	-0.03120	0.02348	-0.03470	0.02503
Common news	-0.00008	0.00059	-0.00211	0.01183

4.3 Sampling and Reference-Window Sensitivity

We also examine the declared grid $\Delta \in \{1, 10, 60\}$ seconds and $W \in \{60, 300, 900\}$ seconds on the same recorded data and observed-flow timing. The cell $\Delta = W = 60$ is excluded because the EMA span is one and its deviation is identically zero. The remaining 24 asset–specification cells use the same gap/session exclusions and date-level summaries. This is exploratory robustness, not immutable external preregistration. Changing the window changes the measured state and its mechanical persistence anchor; it does not merely change the precision of a fixed economic parameter.

The grid should be interpreted as sensitivity of operational diagnostics, not evidence of universal positive feedback. The raw-return comparison and covariance rejections remain relevant even when a filtered-state cell has a positive estimate and a fitted radius below one.

4.4 Covariance Specification Checks

For each asset, fit a seven-parameter linear transition and unrestricted symmetric innovation covariance to 11 lag-0, lag-1 and lag-2 moments. Moment rows exclude all session and gap boundaries. A Bartlett HAC weighting matrix with bandwidth 10 bins is used; the χ^2_4 reference is asymptotic. All three optimizations converge and their numerical moment Jacobians have rank seven, but the overidentifying restrictions are rejected at the 5% reference level.

The rejection can reflect omitted dynamics, filtering, parameter heterogeneity or nonstationarity. The revision does not select a different lag specification solely to obtain acceptance. These results constrain the linear approximation; coefficient sign agreement does not repair misspecification. The prior $J = 73.00$ result belonged to pooled Coinbase data with legacy alignment, not an independent structural validation of the Binance panel.

Identification through heteroskedasticity requires invariant contemporaneous coefficients, orthogonal structural innovations, and nonproportional variance shifts. Those assumptions are not established by grouping volatile and calm observations alone. The previously reported negative contemporaneous product is not used here to identify market-maker buffering or to

Table 3: Fresh sampling and filter grid. Intervals describe date-level variation; radii are maxima over fitted segments. Sign changes and intervals spanning zero are retained.

Asset	Δ	W	EMA anchor	Q [95% interval]	Mean S	Max ρ
BTC	1	60	0.9672	0.00067 [0.00048, 0.00085]	0.9771	0.9886
BTC	1	300	0.9934	0.00031 [0.00022, 0.00040]	0.9952	0.9983
BTC	1	900	0.9978	0.00017 [0.00010, 0.00023]	0.9982	0.9994
BTC	10	60	0.7143	0.00218 [0.00028, 0.00408]	0.7498	0.8174
BTC	10	300	0.9355	0.00194 [0.00032, 0.00355]	0.9426	0.9587
BTC	10	900	0.9780	0.00120 [0.00029, 0.00210]	0.9786	0.9831
BTC	60	300	0.6667	0.00179 [-0.00282, 0.00641]	0.6495	0.7054
BTC	60	900	0.8750	0.00178 [-0.00245, 0.00601]	0.8563	0.8886
ETH	1	60	0.9672	0.00048 [0.00027, 0.00070]	0.9729	0.9838
ETH	1	300	0.9934	0.00028 [0.00018, 0.00038]	0.9942	0.9978
ETH	1	900	0.9978	0.00016 [0.00010, 0.00023]	0.9978	0.9995
ETH	10	60	0.7143	0.00113 [-0.00014, 0.00239]	0.7300	0.7809
ETH	10	300	0.9355	0.00098 [0.00006, 0.00190]	0.9392	0.9574
ETH	10	900	0.9780	0.00072 [0.00009, 0.00136]	0.9781	0.9855
ETH	60	300	0.6667	-0.00271 [-0.00780, 0.00239]	0.6705	0.7168
ETH	60	900	0.8750	-0.00096 [-0.00485, 0.00294]	0.8680	0.8856
ADA	1	60	0.9672	0.00025 [0.00014, 0.00036]	0.9706	0.9796
ADA	1	300	0.9934	0.00014 [0.00008, 0.00019]	0.9935	0.9961
ADA	1	900	0.9978	0.00008 [0.00004, 0.00011]	0.9975	0.9989
ADA	10	60	0.7143	0.00077 [0.00011, 0.00143]	0.7232	0.7462
ADA	10	300	0.9355	0.00062 [0.00016, 0.00108]	0.9350	0.9515
ADA	10	900	0.9780	0.00039 [0.00007, 0.00071]	0.9763	0.9833
ADA	60	300	0.6667	0.00163 [-0.00069, 0.00396]	0.6487	0.7291
ADA	60	900	0.8750	0.00140 [-0.00073, 0.00353]	0.8586	0.8907

Table 4: Fresh covariance-matching diagnostics using observed end-bin flow. Rejection is a model limitation, not validation of structural causality.

Asset	Moment rows	Q_{MD}	J statistic	Reference p
BTC	102454	0.00083	41.41	2.21e-08
ETH	102757	0.00048	28.31	1.08e-05
ADA	102885	0.00091	12.12	0.0165

resolve the rejected dynamic specification.

4.5 Saved Binance Panel: Secondary Evidence

The saved fitted matrices satisfy their recorded Q, S and spectral-radius identities to numerical precision. The initial calculation validated consistency of the derived archive. The subsequent independent raw reconstruction in Section F reproduces all saved core estimates

and supplies corrected observed-time fits; it does not validate the old structural interpretations. Recomputed intervals first average across assets on each of 96 dates and then use a date-level Student-t reference. They remain sensitive to between-date dependence and the legacy forward-flow state definition.

Table 5: Saved Binance derived-panel summaries: 768 asset-days, 96 dates. These are legacy forward-flow diagnostics, not fresh raw-book estimates.

Index	Mean	Date-level 95% interval
Q	0.00815	[0.00730, 0.00901]
S	0.92055	[0.91643, 0.92468]
Q30	0.00640	[0.00550, 0.00730]

The archive has positive Q on 86.85% of asset-days and maximum fitted radius 0.97614. Stationarity of a filtered proxy fit does not certify fundamental price stability or dealer safety. Comparing venues without synchronized joint observations does not establish causal propagation. The previous joint four-variable matrix implied $\rho = 1.00258$, contradicting its reported 0.9612, and no joint generating artifact was established after the expanded workspace and archive search. That old matrix and its associated FEVD percentages are withdrawn. Section E now supplies a new joint fit from restored original-date archives; it does not retrospectively authenticate the withdrawn numbers or identify causal transmission.

The restored original raw panel also permits a corrected observed-flow comparison: mean $Q = 0.000381$, with date interval $[0.000241, 0.000520]$. Its raw-return interval $[-0.000384, 0.000059]$ spans zero. These results, all eight corrected asset means, missingness, and the re-estimated fee contrast appear in Section F. The legacy table above is retained for comparison, not promoted to an observed-time or structural learning estimate.

4.6 July 2022 Fee Intervention

For the preserved 2022 panel, BTCUSDT is the treated pair and seven altcoin USDT pairs are comparators. There are 12 sampled dates before and 12 after July 8, giving 192 asset-day observations. We recompute $Y_{it} = \beta_0 + \beta_1 \mathrm{BTC}_i + \beta_2 \mathrm{Post}_t + \beta_3 \mathrm{BTC}_i \mathrm{Post}_t + e_{it}$ directly from saved estimates. This association is conditional on the legacy measurement construction. The exchange did not randomly assign treatment.

Binance's July announcement also excluded affected BTC volume from VIP and liquidity-provider volume calculations (Binance, 2022a). An August 26–September 26 ETH/BUSD fee promotion could alter cross-pair allocation even though ETHUSDT is the recorded comparator (Binance, 2022b). The intervention bundles trading incentives and may produce

spillovers. Consequently, the treated/control contrast is not an uncontaminated shock to adaptive learning intensity.

Table 6: Recomputed fee-intervention association under alternative covariance/reference assumptions. One treated asset limits all inferential interpretations.

Outcome	Reference	Effect	SE	p	95% interval
Q	Classical OLS	-0.001856	0.003072	0.547	[-0.00792, 0.00420]
Q	Asset clusters, t_7	-0.001856	0.000884	0.074	[-0.00395, 0.00024]
Q	Date clusters, t_{23}	-0.001856	0.003412	0.592	[-0.00891, 0.00520]
S	Classical OLS	-0.000390	0.024585	0.987	[-0.04889, 0.04811]
S	Asset clusters, t_7	-0.000390	0.019526	0.985	[-0.04656, 0.04578]
S	Date clusters, t_{23}	-0.000390	0.013234	0.977	[-0.02777, 0.02699]

The date-clustered feedback interval spans both signs. The effect is not statistically distinguished from zero, but a no-effect conclusion is not established. The cross-asset placebo rank fractions are 0.750 for Q and 1.000 for S; these are descriptive ranks over eight possible treated labels, not randomized-treatment probabilities absent an exchangeability argument. At the illustrative equivalence margin ± 0.007, the date-clustered two-one-sided reference gives $p = 0.073$, so equivalence is not established at 5%. The previous SE of 0.003072 was classical OLS, not asset-clustered uncertainty.

Date-specific BTC-minus-comparator gaps relative to June 24 are saved as descriptive diagnostics. The previous monthly event coefficients and joint pretrend F-tests were not traced to a matching fitted covariance and are withdrawn. The saved event-gap builder borrowed one global SE for all event dates, which does not identify valid date-specific uncertainty. The revision also withdraws the untraced 180% volume headline. A nonsignificant pretrend result, if properly rebuilt later, would still not prove parallel counterfactual trends.

5 Stress Diagnostics and Financial Exposure

5.1 Exploratory Spread-State Regression

On the first 50,000 recorded BTC rows, repeated renders are corrected, timestamps deduplicated, and transitions crossing gaps omitted. A 100-span USD-price EMA defines a dimensional deviation. Regress its next difference on lagged deviation, observed signed base-unit flow, a wide-spread indicator at the sample 90th percentile, and their interactions. The slope-difference covariance uses a 60-second Bartlett HAC estimate. This is an exploratory conditional regression, not a full two-equation structural VAR.

Table 7: Fresh exploratory spread-state diagnostics. Flow slopes are dimensional USD per base-flow unit; they are not the dimensionless loop gain Q.

Diagnostic	Value
Calm flow slope	0.6267
Wide-spread flow slope	0.7510
Slope difference	0.1242
HAC SE of difference	0.2181
Normal-reference p	0.569
Residual SD ratio	1.45
Calm inside-depth mean	152578.07
Wide-spread inside-depth mean	29216.24

Higher residual dispersion and lower measured depth are conditional associations, not evidence of an endogenous instability transition. A slope difference must be interpreted with its uncertainty rather than each regime's separate significance. The old 62.7-fold figure came from a separate 4,999-transition switching-regression fit, whose transition updates and convergence claims were not independently validated here. Neither its extreme p-value nor a dimensionally mixed "stability" index is retained as a headline result. The prior scaled sum added a persistence coefficient to a dimensional slope; this is not a valid unit-invariant boundary.

5.2 Equity Sessions and the January Depth Sample

The event-indexed equity analysis resets OFI and the 100-event EMA at each supplied STOCK–DAY boundary, drops 500 initialization events, and excludes sessions with fewer than 600 raw rows. All reported radii are computed directly. These supplied identifiers and price representations do not support causal comparisons of regulations across venues.

Table 8: Fresh FI-2010 local-representation session fits. STOCK identifiers are supplied metadata; rows differ across stocks and event time is not seconds.

STOCK	Raw rows	Sessions	Mean Q	Mean S	Max ρ
1	18308	9	0.000076	0.97936	0.98810
2	39555	10	0.000295	0.97666	0.98376
3	66862	10	0.000472	0.97683	0.98606
4	71750	10	0.000465	0.97565	0.98099
5	76466	10	0.000594	0.97274	0.98907

The January depth reanalysis streams the entire existing file, carries previous quotes across read chunks, and sums OFI when one second straddles a chunk boundary. It fits a

120-second EMA state against same-second quote OFI, separately by day/gap segment. No executed-trade flow is inferred from book changes.

Table 9: Secondary January 9–20, 2023 depth-sample reanalysis. No claim of prospective holdout validation or authenticated participant attribution.

Diagnostic	Value
Raw depth records	3730870
Dates retained	12
Fitted transitions	844621
Mean Q [date interval]	0.00091 [0.00040, 0.00142]
Mean Q_{30} [date interval]	-0.00130 [-0.00157, -0.00102]
Mean S [date interval]	0.98396 [0.98190, 0.98602]
Maximum segment ρ	0.99294

The 30-lag adjusted feedback changes sign in this sample. Thus a positive baseline is not robust to this adjustment, even though the fitted baseline radii remain below one. This unfavorable sensitivity is retained; quote-update OFI and executed flow are different measurements.

5.3 Fixed-Spectrum Transient Loss

To isolate transient amplification, fix $\rho = 0.85$, $C = [1, 0]$, $B = 1$, and the same Euclidean shock ball in declared abstract coordinates. Vary only the coupling in $J_\theta = \left[\begin{smallmatrix} .85 & \theta \\ 0 & .85 \end{smallmatrix}\right]$. Compute M_C over $t = 0, \ldots, 100$; both nonzero row components decrease beyond their peak, giving a tail certificate for this family. The spectrum is identical in every cell.

Table 10: Fixed-spectrum output magnification. Each exact-radius shock stays at the budget; a constructed worst-direction shock just above it breaches the budget.

Coupling θ	ρ	Peak time	M_C	B/M_C
0.0	0.85	0	1.0000	1.00000
0.5	0.85	6	1.3835	0.72280
2.0	0.85	6	5.3378	0.18734
8.0	0.85	6	21.3012	0.04695

The calculation verifies the norm identity and distinguishes an eigenvalue threshold from a loss-budget threshold. It does not show that arbitrary couplings are economically feasible, or that the initial-shock bound controls repeated stochastic inputs.

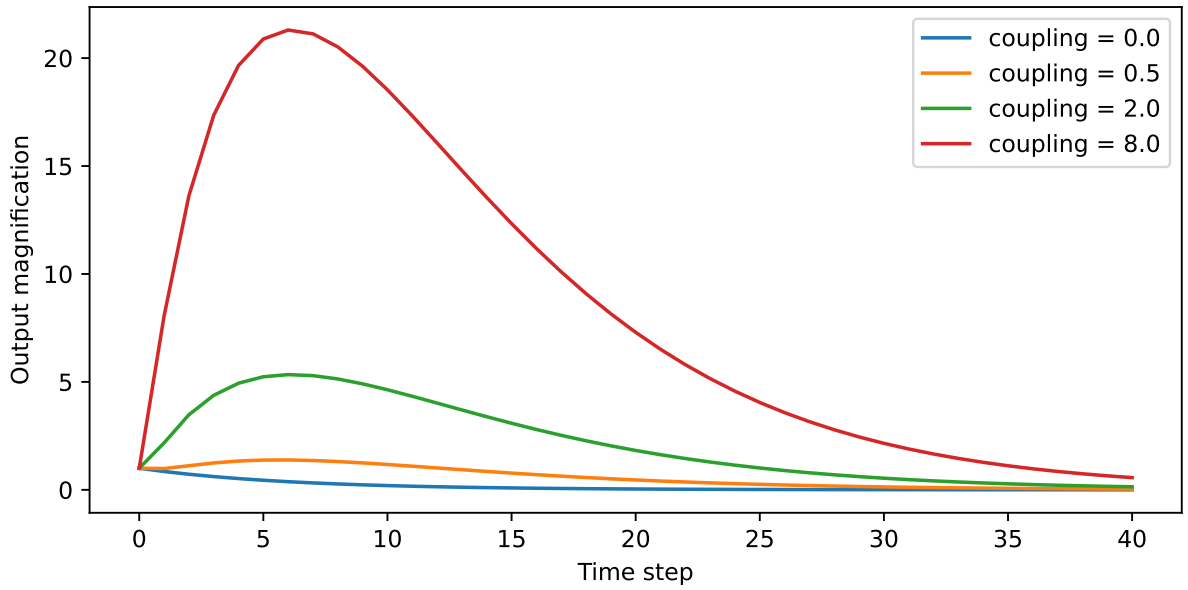


Figure 2: Output magnification with fixed eigenvalues and a fixed loss operator. Coordinates and shock units are held fixed. This numerical family is illustrative rather than market calibrated.

5.4 Funded-Dealer Persistent-Flow Simulation

We separately reproduce 48 cells with 400 seeds each (19,200 paths) using the preserved flow-moment calibration. The model has $a = 0.5$, $\eta = 0.2$, price-impact scale 0.002, price noise 0.0005, spread 0.002, initial equity 1, horizon 500, and inventory control 0.05. The simulation Jacobian uses the lagged update $J = \left[\begin{smallmatrix} a & 1 \\ \eta Q & 1-\eta \end{smallmatrix}\right]$ with $Q = S - a$; its full spectral radius is below one for the reported grid.

The dealer trades against flow at the current price, updates cash by execution proceeds and modeled spread revenue, marks inventory at the next price, and halts when equity is nonpositive or inventory market value exceeds the leverage allowance. These are modeled equity/margin failures, distinct from the historical inventory-capacity experiment, which retained positive equity. A higher allowed leverage limit changes the halt rule; it does not create additional equity.

B1 matches flow variance, B2 matches variance and lag-one autocorrelation, and B3 matches variance and long-run variance with exogenous AR flow. The flow-moment calibration is preserved rather than independently recalibrated from market data. All 48 new seeded halt counts match the recorded counts.

B3 reproduces high failure rates without an endogenous price-to-flow rule. This supports the importance of low-frequency flow exposure within the model; it prevents attributing the failure uniquely to adaptive learning or non-normal transients. The fixed-spectrum

Table 11: Reproduced funded-dealer halt rates, 400 seeds per cell. Exact binomial intervals and halt counts are supplied in the revision results.

Limit L	S	ρ	Feedback	B1	B2	B3
10	0.50	0.8000	11.0%	11.2%	10.5%	10.8%
10	0.60	0.8562	29.5%	11.5%	11.2%	31.5%
10	0.70	0.9000	62.3%	13.0%	14.0%	69.0%
10	0.80	0.9372	93.2%	14.8%	22.2%	97.8%
10	0.90	0.9702	100.0%	26.5%	47.0%	100.0%
10	0.95	0.9854	100.0%	52.8%	87.5%	100.0%
20	0.50	0.8000	0.0%	0.0%	0.0%	0.0%
20	0.60	0.8562	0.0%	0.0%	0.0%	0.0%
20	0.70	0.9000	0.2%	0.0%	0.0%	0.2%
20	0.80	0.9372	5.5%	0.0%	0.0%	10.0%
20	0.90	0.9702	52.5%	0.0%	0.0%	64.5%
20	0.95	0.9854	88.2%	0.0%	3.0%	96.2%

experiment above addresses transient magnification separately. The simulation does not measure actual dealer failures or demonstrate welfare benefits of an exchange gate. Historical flash-crash mitigation percentages are omitted from the main evidence because they were calibrated illustrations rather than new causal event analyses.

6 Attribution, Falsification, and Replication Limits

6.1 Aggregate Relations Do Not Identify Algorithms

The merged-cohort theorem concerns a stated linear, reset observation experiment. Anonymous prices and flows cannot identify identities within a merged representation. More broadly, the empirical VAR here does not distinguish a learning algorithm from a persistent exogenous flow process or discretionary response to common news. Participant identifiers improve disaggregation, but parameter recovery also requires a correct state model, sufficient regressor rank, and an orthogonality assumption; OLS alone is not an automatic causal identification guarantee.

6.2 What the Diagnostics Establish

Timing comparisons show that the legacy and end-bin states differ. Raw-return checks show that filtered persistence is not interchangeable with return persistence. No-adaptation controls test a finite declared family, not every confounder. Covariance rejection remains unfavorable evidence about the low-dimensional approximation. Sign-memory adjustment is

a descriptive robustness exercise, not proof of causal learning. The revision makes no blanket assertion that all placebo tests have passed or that common response proves propagation.

6.3 Traceability and Remaining Limits

The revision archive records hashes, daily estimates, gap handling, inference calculations, numerical checks and source changes. Coinbase and equity calculations use existing local data representations; the January depth file is freshly streamed. The historical Binance results were initially recomputed from saved estimates; Section F now also reproduces them from restored official original-date archives and supplies corrected observed-flow results. Official-source authentication is recorded for the restored Binance inputs. Provider attribution for the recorder, equity and depth representations, immutable external preregistration, independent replication and expert mathematical review remain open. Where numerical provenance could not support the previous claim, the result was withdrawn rather than replaced by a preset favorable number.

7 Conditional Surveillance Implications and Conclusion

7.1 Loss Budgets Require More Than Eigenvalues

The exact radius is conditional on J, fixed state units, a declared shock set, a calibrated loss operator and budget B. If orders enter through a matrix G, the relevant immediate disturbance is $z_0 = Gv$; a scalar price-impact bound does not automatically bound the entire state. For repeated orders or innovations,

$$z_t = J^t z_0 + \sum_{k=0}^{t-1} J^{t-1-k} G w_{k+1},$$

so an initial-shock radius alone is insufficient. A control design would require cumulative exposure bounds from inputs to outputs, and accounting for simultaneous portfolios and uncertain parameters.

7.2 A Conditional One-Burst Gate

If a separately justified model gives $\|z_0\|_2 \le g_t V$ for a burst volume V, then $V \le B_t/(g_t M_C)$ bounds the modeled loss from that burst. The loss budget B_t must come from actual available

capital and declared risk limits, not an unsupported multiplication of capital by leverage. Under $C = I$, $B_t/(en\mathcal{K}(J))$ is a conservative state-shock radius; an upper bound $B_t/\mathcal{K}(J)$ must not be used as a guaranteed safe radius. For a general C, the factor $\|C\|_2$ enters a norm-based conservative bound.

This mathematical implication does not establish that an exchange can observe every required input, run the estimator within its latency budget, improve price discovery, or prevent real insolvency. Comparing execution quality, delayed fills, liquidity and dealer outcomes against ordinary depth/size limits remains a separate prospective study. Circuit breakers and batching address additional market-design questions not resolved by these calculations.

7.3 Conclusion

Observable price–flow data support model-dependent diagnostics, while participant attribution and financial guarantees require additional information and assumptions. Within the stated linear probing experiment, merged cohorts are observationally equivalent and symmetric close-rate alternatives have a fourth-order separation lower bound. The scalar Gaussian price experiment supplies an explicit abstention layer near its unit-root boundary. The loss-budget identity shows why stable eigenvalues alone do not bound worst-case output magnification in fixed coordinates.

Empirically, end-bin observed flow produces smaller Coinbase feedback diagnostics than legacy forward-flow timing; raw-return intervals span zero, and covariance restrictions are rejected. Saved Binance fee contrasts do not establish either a causal learning response or equivalence to zero. Separate fixed-spectrum and funded-dealer simulations distinguish transient magnification from persistent-flow accumulation, with exogenous long-run-variance matching reproducing severe model losses. These limits are part of the findings. The manuscript is a corrected internally audited working paper, with the restored raw panel and joint study reported at their descriptive scope, and provider/structural identification, independent review and real-market policy validation remaining open.

A Full Study Coverage and the Evidence Boundary

This review revision retains all ten study questions and the information-contract and financial-risk program of the original 48-page manuscript. The original submission PDFs are preserved separately. Correcting an invalid result is different from abandoning its scientific question: the evidence below records successful reanalyses, model rejections, and specific missing dependencies. This version is a review candidate with open scientific gates, not a declaration of publication readiness. The page-disposition appendix connects every original page to its revision obligation.

A.1 The Ten-Study Program

Study 1: baseline aggregate diagnostics. The 768 asset-day archive answers a descriptive question about the fitted dynamics of a filtered price/flow proxy. Its matrices and derived invariants are internally consistent, but rechecking a stored matrix is not reproducing the transformation from exchange observations. The historical generator used close-price klines and forward-aligned flow. Accordingly, the archived means remain a separately labeled historical diagnostic. The primary observed-timing reanalysis on the traceable Coinbase sample shows how strongly the same scientific quantity changes under a corrected information convention.

Study 2: cross-asset heterogeneity. Differences in the archived Q means across eight symbols can be reproduced from their saved fits. They do not identify institutional participation or a monotone effect of capitalization, because no participant identities or independently aligned capitalization data enter these estimates. A study of heterogeneity remains useful when sampling, filtering and uncertainty are held fixed. The appendix reports the actual saved means rather than interpreting the rank ordering as a causal mechanism.

Study 3: fee responsiveness. The saved 2022 panel permits reproduction of a treated-minus-control contrast around the fee change. Classical, asset-cluster and date-cluster uncertainty differ substantially. A one-treated-asset comparison needs a defensible counterfactual and assumptions about spillovers. A broad confidence interval containing zero does not establish that fees leave feedback unchanged; the stated equivalence margin also fails under the reported date-cluster test. These are results of the intervention study, not reasons to exclude it.

Study 4: dynamics around the intervention. The twice-monthly dates identify descriptive contrasts relative to the last pre-treatment date. Recovering a date-specific point contrast does not recover its standard error. The inspected event-study generator assigns pooled DiD uncertainty scaled by $\sqrt{2}$, without estimating the covariance of the individual contrast, and its clock differs from the old monthly display. The original joint pretrend statistic is therefore not retained as verified evidence. The dated contrasts remain available below while inference assumptions are resolved.

Study 5: venue transmission. A saved separate-venue comparison is evidence of differences or similarities in marginal fitted dynamics. Transmission requires synchronized joint observations. The old displayed joint matrix contradicts its own spectral-radius claim, and its generating fit was not established in the expanded artifact search. Section E now restores the original overlap from checksum-verified exchange archives and supplies an independently fitted joint process, while retaining clock/missingness sensitivities and the distinction between predictive relations and identified information shocks.

Study 6: forecast-error variance. FEVD additionally requires the residual covariance and a declared innovation identification or generalized decomposition. Percentages cannot be reconstructed from separate marginal summaries. Horizon labels must count the actual bin width: ten fitted steps at ten seconds per step do not represent ten seconds. Ordering sensitivity and the relation between quote midpoints and transaction-close proxies matter. No untraced percentage is offered as an observed systemic contribution. The subsequently restored joint study supplies a complete residual covariance, explicit generalized normalization, and all twenty-four orthogonal orderings; their sensitivity prevents a causal systemic-contribution interpretation.

Study 7: liquidity-state dependence. The first-50,000-record controlled regression is a bounded diagnostic on a recorded sample, not a comprehensive nonlinear identification design. Its slope difference is not detected with HAC uncertainty, while residual scales differ modestly. The historical switching code fits a scalar differenced-EMA regression, uses ad hoc transition updates, and creates a dimensionally chosen index by multiplying a slope by ten. That index is not the structural S in the cohort theorem. A justified regime likelihood, initialization audit and uncertainty are required before a structural bifurcation conclusion.

Study 8: historical stress scenarios. The inspected historical script simulates an E-mini path with imposed depth dynamics and assigns the reported feedback parameters directly.

The Treasury routine returns fixed values. These outputs can motivate an illustrative stress scenario, but they are not estimates from ingested historical TAQ. Moreover, the protected E-mini scenario changes depth and external liquidation flow as well as the proposed interception rule. Its improvement cannot isolate the rule's causal effect. Authenticated historical observation, where recoverable, would be a separate empirical test.

Study 9: the supplied equity sample. Session-reset fits reproduce aggregate diagnostics for supplied STOCK/DAY identifiers. Their counts differ across stocks, and their index is an event clock. The original benchmark/provider map has not been independently authenticated. The estimates therefore support replication of the local calculation, conditional on the supplied representation, rather than a fully authenticated cross-market mechanism. The session audit preserves actual stock counts and excludes cross-session transitions.

Study 10: funded dealers. The 48-cell replay reproduces archived halt counts exactly under preserved flow-moment calibration. Matched independent and persistent-flow comparators are important because inventory accumulation can arise without adaptive learning. Simulation seeds quantify path variability under that calibration; they are not independent market observations. The fixed-coordinate transient examples address a different deterministic experiment. Keeping the two studies separate prevents a simulated halt percentage from being presented as a theorem or historical intervention outcome.

A.2 Information Contracts and Their Distinct Scientific Roles

Public observation, controlled aggregate probing and audited internal state provide different information. A public tape lacks participant identity and policy-state disclosures. An aggregate probe can excite a transfer function under its reset, observation-noise and admissible-input assumptions, but does not automatically separate equal-rate cohorts. Audited cohort-specific state/action histories can remove an aggregation ambiguity in the specified linear model. They cannot on their own establish a correct fundamental-value proxy, a stationary policy, or a calibrated balance-sheet loss mapping.

A regression on disclosed state must specify whether the same-time price innovation enters the state update, which disturbances remain unobserved, and whether measurement error corrupts the disclosed variable. Ordinary least squares consistency cannot be inferred solely from observing a variable called "internal state." Its moment restrictions need independent justification. Similarly, a Kalman recursion is an MMSE statement only under the stated distributional/independence model, and its steady-state gain concerns filtering error rather than financial solvency.

An operational information contract would need an authenticated schema, timestamps, units, missingness policy and verification of access to the declared variables. The present analysis defines conceptual information requirements. It does not claim that a deployed schema implements them or that firms' supplied signals satisfy the model. Uncertainty sets for parameters and output exposures must precede a regulatory action based on a calculated radius. Aggregate model rejection is itself a reason to withhold a structural certification.

B Repeated Innovations: A Conditional Loss Bound

The initial-shock theorem does not control a stream of innovations. This gap has a direct conditional linear-system solution, stated here separately from empirical calibration.

Proposition B.1 (Exact induced gain for bounded innovations)**.** *Let $z_0 = 0$, $z_t = Jz_{t-1} + Dw_t$, $y_t = Cz_t$, and $\rho(J) < 1$. Use fixed Euclidean coordinates and suppose $\|w_t\|_2 \le r$ separately at each time. Define*

$$G = \sup_{t\ge 1} \max_{\|v\|_2=1} \sum_{k=0}^{t-1} \|D^\top (J^k)^\top C^\top v\|_2.$$

Then $G < \infty$ and

$$\sup_{\|w_t\|_2\le r} \sup_{t\ge 1} \|y_t\|_2 = rG.$$

For $G > 0$ and loss budget $B > 0$, the exact radius for this repeated-input experiment is B/G. For a scalar output, $G = \sum_{k=0}^{\infty} \|CJ^kD\|_2$. With an additional initial ball $\|z_0\|_2 \le r_0$, the conservative sufficient condition $r_0 M_C + rG \le B$ controls all times.

Proof. For fixed t, write $y_t = \sum_{k=0}^{t-1} CJ^kDw_{t-k}$. Euclidean duality gives $\|y_t\|_2 = \max_{\|v\|_2=1} v^\top y_t$. The compact finite product of input balls and the unit sphere permits interchanging their maxima. Each input independently maximizes its linear functional to $r\|D^\top (J^k)^\top C^\top v\|_2$, proving the finite-horizon equality. Taking the supremum over t proves the formula. Schur stability implies summability of $\|J^k\|_2$, so $G \le \sum_k \|CJ^kD\|_2 < \infty$. For a scalar output, the dual sphere is $\{-1, 1\}$ and the sum is independent of its sign. The initial-condition inequality follows from the triangle inequality and the definition of M_C. □

This result requires neither nonnegative coupling nor independence of the innovations. It controls bounded adversarial inputs in a fixed linear system; Gaussian innovations are unbounded and do not satisfy its almost-sure premise. A stochastic high-probability result needs a declared horizon and tail law. Applying the proposition to orders requires the actual

input operator D, the economically calibrated loss operator C, fixed units and a justified parameter set. It does not establish market execution safety from a descriptive proxy VAR.

For computation, let G_N be the scalar-output partial sum through $k = N - 1$. Choose an integer h with $q = \|J^h\|_2 < 1$, which exists for a Schur-stable matrix. Commutativity of powers and submultiplicativity give the tail certificate

$$0 \le G - G_N \le \frac{\|D\|_2}{1-q} \sum_{j=0}^{h-1} \|CJ^{N+j}\|_2.$$

Thus a truncated calculation can provide an interval for G rather than assuming that a finite horizon captured its maximum. Near instability the required horizon and uncertainty can be large. For a vector output, the partial dual optimization gives a lower bound and the summed operator norms give a conservative upper bound.

C Measurement and Inference Methods

C.1 The observable transformation

The primary transformation forms past-observed bins, uses the last recorded midpoint in each bin and sums deduplicated executed signed flow. A missing bin splits the sequence; it is not filled from future values. The price state is log midpoint minus its exponential reference with $\alpha = 2/(W/\Delta + 1)$. Five reference horizons are excluded after a segment reset. This changes both the estimand and its sample count. For raw-return diagnostics, differences are formed within the same contiguous segment. Daily aggregation treats multiple segments on one date as parts of one date, rather than independent observations.

The observable proxy is not latent fundamental mispricing. Even a martingale filtered through an EMA produces memory. If a statement concerns fundamental stability, an observation-law relation is needed. Our reduced-form fit includes an intercept, and the off-diagonal ratio defining Q is a diagnostic of that fit. Dividing by $1 - J_{22}$ can amplify uncertainty when flow persistence is close to one; the invariants must be read alongside the full fitted radius, sampling clock and denominator.

C.2 Timing recovery

The recovered Gemini scratch script independently compares one-minute reported buys against sums of one-second records under three interval closures. The deduplicated past interval reproduces every one of the tested 1,995 windows for each asset. The future interval

does not. The corresponding duplicate-render counts are 79,581 (BTC), 73,762 (ETH), and 65,973 (ADA). These arithmetic checks support how to read the local flow timestamps; they do not authenticate the original provider or identify a trader's policy. The revision also reconstructs each full local cache from its raw CSV with exact agreement.

C.3 Dependence and small samples

Means and confidence intervals based on a daily panel describe variation across observed dates. They are not probabilities that an individual latent learning parameter falls in the displayed interval. For the saved Binance panel, all assets on a date are kept together in the date-block summary. The DiD covariance calculations state whether they use classical residual variance, eight asset clusters or twenty-four date clusters, with corresponding reference degrees of freedom. Asset clustering with one treated asset remains fragile, even when the mechanical covariance calculation is correct.

Eight assignment contrasts permit only a coarse descriptive rank fraction unless an exchangeable assignment design is defended. The equivalence margin is a sensitivity choice, not a pre-registered economically optimal tolerance. Failing a TOST or failing to reject a zero effect cannot be combined into evidence that the effect is exactly zero. An event-specific covariance must be calculated for its own contrast, including uncertainty in the reference date.

C.4 Minimum distance and rejection

The reanalysis matches eleven covariance/lag moments to seven parameters and checks local rank seven. Segments contribute their own lagged products; products do not bridge discontinuities. The fitted system is restricted to be stable and innovation covariance positive definite. The diagnostic uses HAC weighting with ten bins and a four-degree-of-freedom asymptotic reference. A successful optimizer supplies a candidate moment fit; rejection assesses the maintained moment model. It cannot establish structural validity through a nearby reduced-form point estimate.

C.5 Null families and specification sensitivity

The nulls are declared no-adaptation data-generating families, not fitted replicas of every feature of the real sample. They isolate independence, persistent exogenous flow, order splitting with impact, and common news. Sixty repetitions per family under the same transformation show that a small positive diagnostic is compatible with mechanisms other

than adaptive learning. This is a bounded falsification exercise; a stronger mechanistic comparison needs calibrated information and uncertainty. The sampling/window grid is exploratory, and all admissible declared cells are retained. The $W = \Delta = 60$ cell is excluded because the chosen EMA becomes degenerate, not because its empirical result is unfavorable.

D Expanded Numerical Ledgers

D.1 Saved cross-asset means

The following legacy means are displayed from the preserved outputs for comparison. Section F independently reproduces them from current official archives and reports corrected observed-time estimates separately. The legacy table retains its original forward-flow construction.

Table 12: Saved legacy panel: all eight asset means.

Symbol	Q	S	Q_{30}
ADAUSDT	0.0067986	0.9143795	0.0061444
BNBUSDT	0.0057950	0.8972684	0.0039697
BTCUSDT	0.0142386	0.9325388	0.0112405
DOGEUSDT	0.0059319	0.9248379	0.0044666
ETHUSDT	0.0131252	0.9338065	0.0106846
LTCUSDT	0.0049256	0.9178751	0.0038416
SOLUSDT	0.0087132	0.9187523	0.0069143
XRPUSDT	0.0057009	0.9249766	0.0039482

D.2 Fee covariance and reference distributions

Table 13: The same DiD coefficient with explicitly different uncertainty.

Outcome	Covariance	Coefficient	SE	df	p
Q	classical	-0.0018555	0.0030720	188	0.54657
Q	asset_cluster	-0.0018555	0.0008841	7	0.07402
Q	date_cluster	-0.0018555	0.0034116	23	0.59176

Outcome	Covariance	Coefficient	SE	df	p
S	classical	-0.0003897	0.0245851	188	0.98737
S	asset_cluster	-0.0003897	0.0195257	7	0.98463
S	date_cluster	-0.0003897	0.0132341	23	0.97676

D.3 Dated event contrasts, without unestimated error bars

Table 14: Descriptive dated contrasts from the same saved 2022 panel; no causal error bars are asserted.

Date	relative_gap_Q
2022-01-10	0.005210
2022-01-24	0.003673
2022-02-10	0.002432
2022-02-24	0.015133
2022-03-10	0.007518
2022-03-24	-0.009571
2022-04-10	-0.000459
2022-04-24	0.005521
2022-05-10	-0.001265
2022-05-24	0.008719
2022-06-10	-0.007705
2022-06-24	0.000000
2022-07-10	0.011556
2022-07-24	0.009562
2022-08-10	-0.001901
2022-08-24	-0.000443
2022-09-10	0.018228
2022-09-24	0.005343
2022-10-10	-0.002254
2022-10-24	-0.007227
2022-11-10	0.003160
2022-11-24	-0.001270
2022-12-10	-0.017021

Date	relative_gap_Q
2022-12-24	-0.010793

D.4 All forty-eight funded-dealer cells

All cells use 400 seeded paths under the archived calibration. Exact binomial intervals concern these conditional path probabilities. They do not quantify calibration error. B_1, B_2, B_3 are the preserved comparator labels; matching halt counts does not imply identical underlying economic mechanisms.

Table 15: Complete conditional dealer replay, including zero and adverse cells.

S	L	Mode	Halts/400	Rate	Exact 95% CI
0.50	10	feedback	44	0.1100	[0.0811, 0.1448]
0.50	10	B1	45	0.1125	[0.0833, 0.1476]
0.50	10	B2	42	0.1050	[0.0767, 0.1393]
0.50	10	B3	43	0.1075	[0.0789, 0.1421]
0.50	20	feedback	0	0.0000	[0.0000, 0.0092]
0.50	20	B1	0	0.0000	[0.0000, 0.0092]
0.50	20	B2	0	0.0000	[0.0000, 0.0092]
0.50	20	B3	0	0.0000	[0.0000, 0.0092]
0.60	10	feedback	118	0.2950	[0.2507, 0.3423]
0.60	10	B1	46	0.1150	[0.0854, 0.1504]
0.60	10	B2	45	0.1125	[0.0833, 0.1476]
0.60	10	B3	126	0.3150	[0.2697, 0.3630]
0.60	20	feedback	0	0.0000	[0.0000, 0.0092]
0.60	20	B1	0	0.0000	[0.0000, 0.0092]
0.60	20	B2	0	0.0000	[0.0000, 0.0092]
0.60	20	B3	0	0.0000	[0.0000, 0.0092]
0.70	10	feedback	249	0.6225	[0.5730, 0.6702]
0.70	10	B1	52	0.1300	[0.0986, 0.1670]
0.70	10	B2	56	0.1400	[0.1075, 0.1779]
0.70	10	B3	276	0.6900	[0.6421, 0.7350]
0.70	20	feedback	1	0.0025	[0.0001, 0.0138]

S	L	Mode	Halts/400	Rate	Exact 95% CI
0.70	20	B1	0	0.0000	[0.0000, 0.0092]
0.70	20	B2	0	0.0000	[0.0000, 0.0092]
0.70	20	B3	1	0.0025	[0.0001, 0.0138]
0.80	10	feedback	373	0.9325	[0.9033, 0.9550]
0.80	10	B1	59	0.1475	[0.1142, 0.1861]
0.80	10	B2	89	0.2225	[0.1827, 0.2665]
0.80	10	B3	391	0.9775	[0.9577, 0.9897]
0.80	20	feedback	22	0.0550	[0.0348, 0.0821]
0.80	20	B1	0	0.0000	[0.0000, 0.0092]
0.80	20	B2	0	0.0000	[0.0000, 0.0092]
0.80	20	B3	40	0.1000	[0.0724, 0.1337]
0.90	10	feedback	400	1.0000	[0.9908, 1.0000]
0.90	10	B1	106	0.2650	[0.2224, 0.3111]
0.90	10	B2	188	0.4700	[0.4202, 0.5202]
0.90	10	B3	400	1.0000	[0.9908, 1.0000]
0.90	20	feedback	210	0.5250	[0.4748, 0.5748]
0.90	20	B1	0	0.0000	[0.0000, 0.0092]
0.90	20	B2	0	0.0000	[0.0000, 0.0092]
0.90	20	B3	258	0.6450	[0.5959, 0.6919]
0.95	10	feedback	400	1.0000	[0.9908, 1.0000]
0.95	10	B1	211	0.5275	[0.4773, 0.5773]
0.95	10	B2	350	0.8750	[0.8385, 0.9058]
0.95	10	B3	400	1.0000	[0.9908, 1.0000]
0.95	20	feedback	353	0.8825	[0.8468, 0.9124]
0.95	20	B1	0	0.0000	[0.0000, 0.0092]
0.95	20	B2	12	0.0300	[0.0156, 0.0518]
0.95	20	B3	385	0.9625	[0.9389, 0.9789]

D.5 Every retained equity session

Table 16: Session-reset equity estimates, including negative session diagnostics. Supplied identifiers are not independently authenticated.

Stock	Day	Raw rows	Transitions	Q	S	ρ
1	1	2127	1626	0.0001870	0.985208	0.985210
1	2	1081	580	-0.0001514	0.979472	0.979468
1	3	2523	2022	-0.0000904	0.979008	0.979006
1	4	2806	2305	0.0000453	0.974563	0.974564
1	5	3030	2529	0.0003415	0.980363	0.980370
1	6	1295	794	0.0004084	0.982068	0.982076
1	8	1493	992	-0.0012502	0.975850	0.975814
1	9	1873	1372	0.0008702	0.988085	0.988096
1	10	1888	1387	0.0003243	0.969608	0.969618
2	1	1327	826	0.0004680	0.983753	0.983761
2	2	3998	3497	0.0000513	0.981441	0.981442
2	3	1380	879	0.0003006	0.960802	0.960811
2	4	6992	6491	0.0007136	0.981525	0.981539
2	5	6728	6227	0.0001719	0.975830	0.975834
2	6	968	467	0.0000588	0.975327	0.975328
2	7	2609	2108	0.0002809	0.978361	0.978367
2	8	1154	653	0.0000752	0.975439	0.975440
2	9	9271	8770	0.0003781	0.977952	0.977961
2	10	5128	4627	0.0004528	0.976125	0.976135
3	1	6318	5817	0.0005351	0.978800	0.978812
3	2	6122	5621	0.0006917	0.986049	0.986059
3	3	5411	4910	0.0002967	0.970929	0.970937
3	4	5342	4841	0.0009275	0.984479	0.984495
3	5	5946	5445	0.0008630	0.982060	0.982076
3	6	6243	5742	0.0004446	0.979903	0.979912
3	7	7060	6559	0.0005057	0.976492	0.976504
3	8	8662	8161	0.0000745	0.978591	0.978593
3	9	10036	9535	0.0001343	0.972203	0.972207
3	10	5722	5221	0.0002512	0.958831	0.958842
4	1	4922	4421	0.0006913	0.980980	0.980994

Stock	Day	Raw rows	Transitions	Q	S	ρ
4	2	5965	5464	0.0006941	0.975722	0.975740
4	3	4027	3526	0.0005459	0.978853	0.978865
4	4	10819	10318	0.0003697	0.977566	0.977574
4	5	6378	5877	0.0004720	0.979251	0.979261
4	6	5607	5106	0.0008354	0.980852	0.980869
4	7	6740	6239	0.0001300	0.960876	0.960881
4	8	8591	8090	0.0003745	0.973590	0.973600
4	9	12880	12379	0.0001835	0.976114	0.976118
4	10	5821	5320	0.0003553	0.972704	0.972714
5	1	10719	10218	0.0000036	0.976763	0.976763
5	2	6712	6211	0.0002562	0.977963	0.977969
5	3	5186	4685	0.0003102	0.973532	0.973540
5	4	8223	7722	0.0002691	0.989064	0.989067
5	5	1865	1364	0.0008953	0.983205	0.983220
5	6	7007	6506	0.0006622	0.982051	0.982063
5	7	7854	7353	0.0001806	0.980382	0.980386
5	8	13229	12728	0.0001101	0.975979	0.975981
5	9	14973	14472	0.0000975	0.980657	0.980659
5	10	698	197	0.0031508	0.907759	0.908027

E Restored Joint-Venue Study

The original separate-venue generator did not establish its reported joint matrix or FEVD. That contradiction remains in the audit. A subsequent recovery retrieved the exact original study's Binance BTCUSDT, ETHUSDT and ADAUSDT daily one-second kline objects for April 8–18, 2021: thirty-three exchange archives, each verified against a newly retrieved official checksum and its ZIP CRC. These are current public archives for the original sample dates, not byte-authenticated copies of the original downloads. Three recovered April 10 asset-days reproduce every saved baseline coefficient, invariant, observation count and mean price exactly. Full original-panel recovery and reproduction are recorded separately; this three-row overlap comparison alone does not close that full-panel gate.

The new independent joint study joins the three recorder caches to restored exchange observations. Binance supplies transaction closes and signed quote flow $2V^{\text{buy}} - V^{\text{total}}$, rather than order-book midpoints. Coinbase supplies recorded midpoint and deduplicated

flow, with provider authentication still open. Quote currencies differ (USD and USDT); basis movements and common information can affect the relation. The two timestamp conventions do not establish subsecond simultaneity. We retain integer-second offset sensitivities of minus and plus one second for the recorder.

Ten-second bins require all ten integer-second records from both venues. Incomplete bins are excluded; neither series is filled from future observations. Gaps and UTC date changes reset both filters together, followed by a 150-bin burn-in. Segments require at least 500 retained bins. A four-variable VAR with separate segment intercepts is fit to the within-segment centered observations. The ordered state is recorder price deviation, recorder flow, Binance price deviation, Binance flow. Each variable is divided by its retained-sample standard deviation for estimation; the saved fit also supplies the raw-unit matrix and covariance. The filtered state uses a 300-second EMA; a raw-return variant measures a different estimand.

Table 17: Joint fits from restored original overlap. Radii describe fitted reduced-form dynamics, not fundamental stability.

Asset	State / clock	Transitions	Segments	ρ
BTC	EMA, offset 0	85304	64	0.950614
BTC	Return, offset 0	85304	64	0.286970
BTC	EMA, offset -1s	85302	64	0.952560
BTC	EMA, offset +1s	85303	64	0.950148
ETH	EMA, offset 0	85008	66	0.949934
ETH	Return, offset 0	85008	66	0.383452
ETH	EMA, offset -1s	85008	66	0.949761
ETH	EMA, offset +1s	85008	66	0.950104
ADA	EMA, offset 0	85615	62	0.948204
ADA	Return, offset 0	85615	62	0.227168
ADA	EMA, offset -1s	85616	62	0.946539
ADA	EMA, offset +1s	85617	62	0.948877

For horizon H, with $\Psi_k = J^k$ and residual covariance Σ, the generalized decomposition (Pesaran and Shin, 1998) is

$$\theta_{ij}(H) = \frac{\Sigma_{jj}^{-1} \sum_{k=0}^{H-1} (e_i^\top \Psi_k \Sigma e_j)^2}{\sum_{k=0}^{H-1} e_i^\top \Psi_k \Sigma \Psi_k^\top e_i}.$$

With correlated innovations, these raw generalized shares need not sum to one. We report $\widetilde{\theta}_{ij} = \theta_{ij} / \sum_j \theta_{ij}$ as an explicitly chosen row normalization, retain raw shares in the output, and do not interpret their directional difference as identified net causality. We also calculate orthogonal decompositions under all twenty-four variable orderings. Six fitted steps are

sixty seconds. The matrix, full residual covariance, responses, and every horizon use the same saved joint fit.

Table 18: Sixty-second cross-price predictive decomposition, percentages. Ordering ranges are sensitivity ranges, not confidence intervals.

Asset	BN to CB	Order range	CB to BN	Order range
BTC	44.29	[17.00, 84.83]	32.24	[0.05, 57.77]
ETH	41.31	[4.77, 79.46]	37.45	[0.73, 69.99]
ADA	32.52	[0.70, 60.87]	30.75	[0.20, 56.51]

The ordering ranges are broad. Contemporaneously correlated innovations can produce cross-price shares even without a directed lag channel. These calculations establish a reproducible joint descriptive study; they do not identify a structural information shock, eliminate common news, or validate the old untraced percentages. Date-resampling with 200 draws is supplied as exploratory variation under an exchangeable-date assumption, not as a small-sample causal guarantee. Residual lag correlations are retained; fitting a stable matrix does not establish model adequacy. The raw-return and clock-offset fits remain visible rather than selecting a preferred result.

F Raw Reproduction of the Original Binance Panel

All 768 original asset-day inputs are now restored from current official exchange archives, with their checksums and ZIP CRC verified. They contain 66,316,704 CSV rows and 66,316,341 unique seconds; the legacy filled fits contain 6,519,552 states. The corrected observed-time analysis retains 6,513,239 transitions in 784 contiguous segments. Every saved baseline coefficient, invariant, mean price, state count, and flow-controlled coefficient is reproduced within the stated numerical tolerance. The legacy loop ratio is independently recomputed from its defining identity. The old covariance-matching and heteroskedastic structural interpretations are not certified by these numerical matches. Original download-time hashes were not preserved; the receipt anchors the current official input snapshot.

Across the original dates, 38,859 seconds are missing and 363 timestamp records are exact duplicates, concentrated on December 24, 2021 and March 24, 2023. No conflicting duplicate values were found. The legacy forward-fill and zero-flow substitution are retained only to reconstruct the saved calculations. Corrected bins require ten recorded seconds, reset at gaps and dates, and exclude five EMA spans after each reset. Segment estimates are combined within dates using transition counts. The resulting estimand differs from the legacy forward-flow alignment.

Table 19: Full-precision fits are saved digitally; displayed normalized matrices round to six decimals. Row/column order is CB price, CB flow, BN price, BN flow.

Asset	Matrix	Row	CB price	CB flow	BN price	BN flow
BTC	J	1	0.615681	0.017169	0.344680	0.016968
BTC	J	2	-0.281880	0.173033	0.437549	0.036387
BTC	J	3	0.025606	0.016819	0.919871	0.009472
BTC	J	4	0.037385	0.032498	0.026393	0.156420
BTC	Σ	1	0.077404	0.096316	0.062844	0.069130
BTC	Σ	2	0.096316	0.911571	0.074506	0.159400
BTC	Σ	3	0.062844	0.074506	0.093297	0.095180
BTC	Σ	4	0.069130	0.159400	0.095180	0.959249
ETH	J	1	0.775163	0.003957	0.176204	0.020426
ETH	J	2	-0.147239	0.139105	0.301382	0.046338
ETH	J	3	0.073362	0.008001	0.872432	0.011210
ETH	J	4	0.048283	0.039717	0.017520	0.080380
ETH	Σ	1	0.095445	0.083451	0.077468	0.073354
ETH	Σ	2	0.083451	0.930111	0.065504	0.134739
ETH	Σ	3	0.077468	0.065504	0.099874	0.096972
ETH	Σ	4	0.073354	0.134739	0.096972	0.981444
ADA	J	1	0.893862	0.007220	0.050108	0.021432
ADA	J	2	-0.049199	0.120553	0.128009	0.066042
ADA	J	3	0.024676	0.002544	0.922499	0.012366
ADA	J	4	0.064629	0.013664	0.025311	0.099010
ADA	Σ	1	0.107064	0.097674	0.075204	0.111943
ADA	Σ	2	0.097674	0.961497	0.064103	0.176227
ADA	Σ	3	0.075204	0.064103	0.100421	0.139113
ADA	Σ	4	0.111943	0.176227	0.139113	0.974483

Table 20: Full original panel, three measurement constructions. Intervals describe variation across 96 date means; positive-share percentages count asset-days.

Construction	Q [date interval]	Q_{30} [date interval]	Mean S	Positive Q (%)	Max ρ
Legacy EMA / forward	0.008154 [0.007296, 0.009011]	0.006401 [0.005503, 0.007299]	0.920554	86.85	0.976140
EMA / observed	0.000381 [0.000241, 0.000520]	0.000285 [0.000158, 0.000411]	0.920383	62.37	0.974559
Return / observed	-0.000163 [-0.000384, 0.000059]	–	-0.041472	48.83	0.465670

Corrected observed-flow feedback is much smaller than its reproduced legacy counterpart. Raw-return intervals span zero. Circular date-block bootstrap sensitivities with block lengths two, four and eight sampled dates are retained in the digital results; they do not establish stationarity, serial independence or causal learning. The sign share is not a participant-attribution statistic.

The original sampled-day BTCUSDT quote-volume mean rises from 2,543,076,971.10 before the fee change to 4,450,660,830.53 after it, a measured 75.01% change in the recovered representation, rather than the old uncomputed 180% literal. This is a descriptive com-

Table 21: All eight assets under corrected observed-flow timing, with raw-return comparison. These are reduced-form diagnostic means, not identified learning gains.

Asset	Observed Q	Observed Q_{30}	Return Q	Observed S
ADAUSDT	0.0003953	0.0003623	0.0000840	0.914243
BNBUSDT	0.0004587	0.0002731	-0.0000789	0.894655
BTCUSDT	0.0006350	0.0005751	0.0000594	0.934638
DOGEUSDT	0.0002511	0.0001622	-0.0002345	0.924580
ETHUSDT	0.0005657	0.0003358	-0.0005013	0.934915
LTCUSDT	0.0003068	0.0002650	0.0003200	0.917590
SOLUSDT	0.0002036	0.0001314	-0.0005210	0.919080
XRPUSDT	0.0002292	0.0001741	-0.0004279	0.923359

Table 22: Fee-period association recalculated on the raw-reproduced, corrected observed-flow panel. Reference covariance assumptions remain fragile with one treated asset.

Outcome	Covariance	Effect	SE	p	95% interval
Q	Classical	-0.0001057	0.0004038	0.7938	[-0.0009023, 0.0006909]
Q	Asset clusters, t_7	-0.0001057	0.0002139	0.6363	[-0.0006116, 0.0004001]
Q	Date clusters, t_{23}	-0.0001057	0.0004449	0.8143	[-0.0010260, 0.0008146]
S	Classical	-0.0036464	0.0254227	0.8861	[-0.0537968, 0.0465039]
S	Asset clusters, t_7	-0.0036464	0.0203732	0.8630	[-0.0518215, 0.0445286]
S	Date clusters, t_{23}	-0.0036464	0.0140420	0.7974	[-0.0326946, 0.0254017]

parison of the original sampled dates; price levels, market trends, bundled incentives and spillovers prevent a fee-induced volume interpretation.

At the inherited illustrative equivalence margin ± 0.007, the corrected diagnostic's date-cluster TOST gives $p = 5.76e - 14$. This margin is over eighteen times the corrected mean Q, so passing that reference test is not evidence of economically small effects or no causal response. The date-cluster effect interval remains compatible with changes large relative to the corrected baseline. The legacy estimator's failed equivalence test concerns a different measurement construction and remains visible.

G Why the Observation Contract Changes the Abstention Result

The scalar price-surveillance theorem cannot be promoted into a mandatory-abstention theorem for every richer observation experiment. The following constructive result resolves a specific full-observation case rather than leaving that distinction informal.

Proposition G.1 (Two-step identification under an exact audited contract)**.** *Consider the single-cohort recursion*

$$x_t = ax_{t-1} + \kappa u_{t-1} + w_t, \qquad u_t = (1-\eta)u_{t-1} + \eta q x_t,$$

with aggregate cohort demand u_t, $x_0 = u_0 = 0$, known $a \in [0,1)$ and $\kappa > 0$, unknown $\eta \in (0,1)$ and $q \in \mathbb{R}$, and w_1 has a continuous nondegenerate Gaussian law. Suppose the true pricing error x_t, cohort demand u_t, and their exact timing are observed without measurement error for $t = 1, 2$. Then, with probability one, either $q = 0$ is identified and the model is Schur stable, or both q and η are recovered exactly:

$$\vartheta = \frac{u_1}{x_1}, \qquad \beta = \frac{u_2 - \vartheta x_2}{u_1}, \qquad \eta = 1 - \beta, \qquad q = \frac{\vartheta}{\eta}.$$

Consequently the exact signed-feedback Jury conditions decide stability with zero error in this observation experiment, including alternatives arbitrarily close to its stability boundaries.

Proof. The reset gives $x_1 = w_1$, so $x_1 \neq 0$ almost surely. The first demand observation gives $u_1 = \eta q x_1$. If $u_1 = 0$, then $q = 0$ because $\eta > 0$; the triangular zero-feedback dynamics have eigenvalues a and $1 - \eta$, both strictly inside the unit circle. Otherwise $u_1 \neq 0$ and $\vartheta = \eta q$. The second demand equation gives $u_2 - \vartheta x_2 = (1-\eta)u_1$, yielding the recovery formulas. With $Q = \kappa q$, the signed Jury interval is

$$-\frac{(2-\eta)(1+a)}{\eta} < Q < 1 - a.$$

All parameters in this decision are known or exactly recovered. Equality is a unit-circle boundary and is classified as not Schur stable. Process innovations do not obstruct recovery because the exact same-time price appears in the observed demand recursion. □

This is an exact mathematical information contract, not a claim that public midpoint records reveal fundamental mispricing or that a dealer supplies noiseless policy states. Timing errors, unknown state definitions, additional cohorts, discretionary orders, noisy disclosure or unknown a, κ change the experiment. Numerical floating-point decoding does not inherit an exact boundary decision without an error analysis. The result demonstrates why a price-only lower bound and an audited-state guarantee need separate assumptions; it does not undo the scalar Gaussian lower bound.

Proposition G.2 (Bounded merged-cohort identification under exact disclosure)**.** *Let $x_0 = y_{i,0} = 0$, $x_t = ax_{t-1} + \kappa u_{t-1} + w_t$, $y_{i,t} = (1-\eta_i)y_{i,t-1} + \eta_i x_t$, and $u_t = \sum_i q_i y_{i,t}$, with known*

$a \in [0,1)$ and $\kappa > 0$, rates $\eta_i \in (0,1)$, and finite signed weights. Suppose at most $K_{\max} \geq 1$ distinct rates have nonzero merged weights. Under exact observation of true x_t and cohort demand u_t for $t = 1, \ldots, 2K_{\max}$ and a continuous nondegenerate Gaussian first innovation, the merged rate–weight measure is identified almost surely. This identifies the closed-loop stability class under the maintained model, although participant counts within merged groups remain unidentified.

Proof. Iterating the reset cohort recursion gives

$$u_t = \sum_{k=0}^{t-1} h_k x_{t-k}, \qquad h_k = \sum_{j=1}^{K} c_j r_j^k, \quad r_j = 1 - \eta_j \in (0,1), \quad c_j = \eta_j \bar{q}_j \neq 0,$$

where $K \leq K_{\max}$ counts nonzero merged groups. Since $x_1 = w_1 \neq 0$ almost surely, the first $2K_{\max}$ moments are recovered recursively:

$$h_n = \frac{u_{n+1} - \sum_{k=0}^{n-1} h_k x_{n+1-k}}{x_1}, \qquad n = 0, \ldots, 2K_{\max} - 1.$$

For uniqueness, the difference of two candidate signed measures has at most $2K_{\max}$ distinct nodes. Vanishing first $2K_{\max}$ moments implies zero weights by the full column rank of the corresponding Vandermonde matrix. Thus the merged measure is unique, including signed weights.

Constructively, the $K_{\max} \times K_{\max}$ Hankel matrix $H = (h_{i+j})_{i,j=0}^{K_{\max}-1}$ factors as $V \operatorname{diag}(c_j) V^\top$, where $V_{ij} = r_j^i$. Its rank is K; the leading $K \times K$ block is invertible. Solve

$$\sum_{j=0}^{K-1} h_{i+j} b_j = -h_{i+K}, \qquad i = 0, \ldots, K-1.$$

The roots of $r^K + \sum_{j=0}^{K-1} b_j r^j$ are the distinct memory nodes. The first K moments then determine the c_j through a square Vandermonde system, and $\bar{q}_j = c_j/(1-r_j)$. When $H = 0$, the merged measure is empty. The identified merged groups determine the minimal closed-loop Jacobian. Additional same-rate difference modes and canceled groups have eigenvalues $r_j \in (0,1)$, so they do not change its instability classification. □

The constructive step uses established finite exponential-sum and annihilating-filter algebra, rather than a new recovery algorithm; see, for example, Vetterli et al. (2002). The contribution here is the explicit observation-contract comparison for the stated market recursion. Exact algebra does not imply robust finite-precision recovery: close rates, small weights, noisy demand, and small x_1 can make inversion ill-conditioned. With no bound on merged model order, this finite observation budget does not supply the same guarantee. Anonymous

recorded prices and total executed flow do not certify this exact true-state/cohort-demand contract.

H Recovering Evidence Across Workspaces

The search receipt indexes 14,804 text files and archive members, with 10,570 distinct content hashes. It includes related local release/repair folders, historical tar releases, a Zoom ZIP and Gemini/Antigravity scratch material. Texts larger than two megabytes, environments, credentials and Git objects were excluded from content matching. This is a bounded discovery search rather than a declaration that all possible backups have been read. Candidate text matches are reviewed semantically before they support a result.

The recovered numeric-provenance matcher accepts a number if some token anywhere in a JSON/CSV rounds to it, including sign changes and factors of one hundred. Such a rule can accept an unrelated observation or simulated input and does not establish a generating chain. A source-to-data-to-estimator-to-field link is required instead. A subsequent current-source ledger checks 1,694 cells across all twenty-three main and appendix tables against their specifically named result fields, with no rendering mismatch. It separately records nine input hashes and inventories theorem, figure and numerical-prose review obligations. These checks do not certify the scientific assumptions, provider identity, causal interpretation, or novelty. Agent completion reports and test counts are useful retrieval leads, but do not independently authenticate a historical sample or a theorem.

Additional named data and release directories were listed on the external T9 volume, including January 2025 archives. After a temporary loss of access, the volume became readable and 458 research files were inventoried, including 192 data archives with matching checksum-file names. Small research JSON outputs were read and hashed. A subsequent recovery audit streamed all 192 already located compressed data archives (6,510,779,702 bytes) and matched each SHA-256 digest to its supplied local checksum. Archive member names and initial CSV records were inspected without extracting new datasets. This establishes payload integrity relative to those checksum records, not independent provider authentication or complete CSV validation. The named binance-one-second directory contains no data files. The located archive dates and types do not establish the older one-second panel. An older January report repeats a withdrawn Coinbase value of -0.3796 and has conclusion prose inconsistent with its own positive estimates; recovery does not resolve those scientific conflicts. The subsequent recovery restores only original study units, as detailed above. No additional sample expansion or complex learner training was launched to substitute for evidence repair.

I Disposition of Every Original Page

The table below refers to the original 48-page submission PDF, not the pagination of this review revision. It retains every original topic. Items marked open require the precise dependencies specified in the accompanying gap register, rather than silent removal from scope. Completed local calculations and unresolved scientific identification are deliberately distinguished.

Table 23: Full original-page disposition.

Page	Original topic	Revision obligation
1	Title/abstract	Replace causal/adaptive and market-validation claims with supported experiment-specific claims. Keep the three scientific questions.
2	Motivation and delegation claims	Distinguish motivating conjectures from measured AI adoption; add precise evidence for temporal/market assertions.
3	Contribution roadmap	Retain cohort identification, surveillance limits, empirical response and financial safety; map each to conditional theorem or descriptive evidence.
4	Fee and core empirical headlines	Use corrected timing and fee inference; retain unfavorable results and distinguish saved panel from raw reproduction.
5	Related literature	Compare exact claims with microstructure, identification and nonnormal financial literature; no broad first-use assertion.
6	Funding-liquidity and transient literature	Explain the added financial output mapping and its limits; keep established prior transient work.
7	Ten-study program	Keep all ten original study questions. Mark cross-venue/FEVD open and historical scenarios simulated; do not count them as empirical replication.
8	Organization/model setup	Align the complete program with results, including unresolved studies. State model assumptions before theorems.
9	Pricing equation and fundamental state	Separate latent fundamental deviation from observed EMA/close proxy and specify disturbance law.

Page	Original topic	Revision obligation
10	Online learner derivation	Retain derivation with H>0 and step/leak constraints; it is a stylized learner, not proof about all algorithms.
11	Single-cohort stability boundary	Keep exact positive-feedback conditions and recompute signed/lagged Jury conditions before extension.
12	Cohort transfer equivalence/minimal realization	Specify zero initial state, process noise, known parameters, nonzero merged weights and no cancellation.
13	Near-cohort separation proof	Use bounded compact parameter class, fixed nonzero signal/noise and valid contour/separation constants.
14	Finite-window testing theorem	Replace unproved full-observation reduction with proved scalar experiment, and record full economy as open theorem.
15	Testing likelihood/ abstention proof	Use exact finite-window KL and total variation bounds for the stated scalar Gaussian law.
16	Abstention decision figure	Generate an illustrative decision curve only with declared rule/experiment; do not claim universal mandatory abstention on every path.
17	Financial loss operator/budget	Use fixed-coordinate C, norm and B as loss allocation from equity; specify initial-shock deterministic experiment.
18	Worst-case radius and Kreiss bounds	Keep exact B/M_C; preserve ||C|| factors and correctly distinguish upper/lower bounds.
19	Jordan construction	Correct discrete maximizer; fresh fixed-rho examples show peak6 for theta=.5,2,8 and computed radii.
20	Pseudospectrum illustration	Retain analytical connection; replace uncomputed empirical-looking contour with reproducibly computed example if desired.
21	Directed cross-asset coupling	Retain proof under nonnegative coupling and nonnegative C; general signed claims remain open.
22	Reduced-form/ structural parameter mapping	Call proxy VAR descriptive and list conditions required for structural mapping; no trader attribution.
23	Flow memory controls	Keep Q30 control and dependence diagnostics; January controlled coefficient is negative and must appear.

Page	Original topic	Revision obligation
24	Minimum distance and heteroskedastic identification	Report fresh rejection tests with actual HAC/rank and separately audit causal variance-shift conditions.
25	Book/flow data description	Label midpoint snapshots, executed notional and quote OFI accurately; Binance close bars are not L3.
26	Cleaning/ synchronization protocol	Use exact cache reconstruction, duplicate removal, past-flow bins and gap/session resets; provider authentication remains open.
27	Headline table and pooled panel	Regenerate from saved matrices with date-block intervals, correct counts and positivity86.85%, not99.7%.
28	Structural validation results	Do not describe J rejection as model validation; include every fresh asset p-value and inferential scope.
29	Cross-asset heterogeneity table	Regenerate saved-panel asset means/uncertainty and disclose legacy timing; raw reproducibility remains separate.
30	Fee design and economic mechanism	Document concurrent fee/volume changes and contamination possibilities; no unidentified no-effect conclusion.
31	DiD table/placebo/ equivalence	Regenerate covariance-specific df,p,CI; TOST at $\pm.007$ does not establish equivalence at5%.
32	Dynamic event study/ pretrend test	Keep study topic, but replace untraced monthly statistics with traced descriptive date gaps pending valid uncertainty.
33	Joint cross-venue VAR table	Record exact eigenvalue contradiction; recover raw join and refit before propagation claims.
34	FEVD/lead-lag table	No FEVD inference from separate venue summary; require joint fit, residual covariance, ordering and clock.
35	Liquidity threshold and switching regimes	Show corrected controlled/HAC diagnostic; archive old scalar switching model with explicit defects, no fabricated extremep.
36	2010/2014 historical stress interpretation	Distinguish official historical descriptions from this script's simulated paths and assigned feedback.
37	Equity replication table	Use actual unequal stock counts, reset sessions and new estimates; authenticate representation before named-provider claims.

Page	Original topic	Revision obligation
38	Funded dealer mechanism/table	48-cell replay reproduces all archived halt counts, conditional on existing moment calibration.
39	Financial interpretation of halt results	Retain calibration sensitivity and differences between one-off radius and repeated innovations.
40	Dealer phase diagram	Provide seed denominators and exact binomial intervals; neither spectral stability nor simulated halts certify real-market safety.
41	Information contract architecture	Keep I0/I1/I2 question and specify observables/disclosures; no implementation guarantee from conceptual architecture.
42	Audited state regressions and Kalman claims	Check noise independence/Gaussian and disclosed-state conditions; distinguish architecture access from identified economic mapping.
43	Falsification battery	Trace each old statistic, include rejected MD and mechanical nulls, avoid literal-presence tests as empirical validation.
44	Sampling/memory sensitivity	Fresh24 cells retain negative estimates and CIs spanningzero; anchor depends strongly on filter.
45	January holdout/ table	Use actual Jan9–20 coverage and negativeQ30; remove untouched/fullmonth label absent freeze provenance.
46	Regulatory consequence/conclusion	Restore complete unclipped conclusion; use correct loss units and conditional safety statement; disclose unresolved studies.
47	References part1	Check cited claims and metadata, especially theorem sources, intervention dates and dataset/provider representation.
48	References part2	Complete bibliography and direct primary-source records; add data/provenance/evidence appendix links.

J Remaining Gates and the User-Requested Deferrals

Raw reconstruction of all 768 original panel units and an independent synchronized joint study are now complete at their declared descriptive level. The active work remains full claim-ledger closure, residual/model diagnostics, existing policy/manifest verification,

provider authentication and independent review. Neither the reconstructed panel nor its predictive decomposition identifies adaptive learning or a causal venue shock. Provider authentication, observational causal identification and independent mathematical/econometric review cannot be replaced by a compilation or test pass.

New large datasets, new complex learning-agent experiments and extra market modules are deferred until existing evidence is reliable. The currently located datasets and existing experiment outputs remain in the recovery/audit scope. T9 metadata and small-result checks were reopened when the volume became readable; original panel inputs have now been restored from current official archives and their saved core estimates reproduced; original download-time bytes and broader structural interpretations are not certified. Historical TAQ estimation reopens when an authenticated existing source is recovered; illustrative simulation parameters do not fill that gap. External replication and journal submission follow a concrete reproducible package, not an agent declaration of completion.

A rejection, null effect or invalid interpretation is already a result and must be reported now. These unfavorable findings are not deferred. The full revision retains their scientific questions while correcting their evidentiary status.

References

Adrian, T., and H. S. Shin, 2010, Liquidity and leverage, *Journal of Financial Intermediation* 19, 418–437.

Almgren, R., and N. Chriss, 2000, Optimal execution of portfolio transactions, *Journal of Risk* 3, 5–39.

Andrews, D. W., 1991, Heteroskedasticity and autocorrelation consistent covariance matrix estimation, *Econometrica* 59, 817–858.

Biais, B., T. Foucault, and S. Moinas, 2015, Equilibrium fast trading, *Journal of Financial Economics* 116, 292–313.

Bouchaud, J.-P., Y. Gefen, M. Potters, and M. Wyart, 2004, Fluctuations and response in financial markets: The subtle nature of "random" price changes, *Quantitative Finance* 4, 176–185.

Brunnermeier, M. K., and L. H. Pedersen, 2009, Market liquidity and funding liquidity, *Review of Financial Studies* 22, 2201–2238.

Budish, E., P. Cramton, and J. Shim, 2015, The high-frequency trading arms race: Frequent batch auctions as a market design response, *Quarterly Journal of Economics* 130, 1547–1621.

Campbell, J. Y., S. J. Grossman, and J. Wang, 1993, Trading volume and serial correlation in stock returns, *Quarterly Journal of Economics* 108, 905–939.

CFTC and SEC, 2010, Findings regarding the market events of May 6, 2010: Report of the staffs of the CFTC and SEC to the Joint Advisory Committee on Emerging Regulatory Issues, Technical report, Commodity Futures Trading Commission and Securities and Exchange Commission.

Cont, R., S. Stoikov, and R. Talreja, 2010, A stochastic model for order book dynamics, *Operations Research* 58, 549–563.

Cont, R., A. Kukanov, and S. Stoikov, 2014, The price impact of order book events, *Journal of Financial Econometrics* 12, 47–88.

Donald, S. G., and K. Lang, 2007, Inference with difference-in-differences and other panel data, *Review of Economics and Statistics* 89, 221–233.

Eisenberg, L., and T. H. Noe, 2001, Systemic risk in financial systems, *Management Science* 47, 236–249.

Gilbert, E. G., 1963, Controllability and observability in multivariable control systems, *Journal of the Society for Industrial and Applied Mathematics, Series A: Control* 1, 128–151.

Glosten, L. R., and P. R. Milgrom, 1985, Bid, ask and transaction prices in a specialist market with heterogeneously informed traders, *Journal of Financial Economics* 14, 71–100.

Greenwood, R., A. Landier, and D. Thesmar, 2015, Vulnerable banks, *Journal of Financial Economics* 115, 471–485.

Hasbrouck, J., 1991, Measuring the information content of stock trades, *Journal of Finance* 46, 179–207.

He, Z., and A. Krishnamurthy, 2013, Intermediary asset pricing, *American Economic Review* 103, 732–773.

Hendershott, T., C. M. Jones, and A. J. Menkveld, 2011, Does algorithmic trading improve liquidity?, *Journal of Finance* 66, 1–33.

Joint Treasury Staff, 2015, The U.S. Treasury market on October 15, 2014, Staff report, U.S. Department of the Treasury, Board of Governors of the Federal Reserve System, Federal Reserve Bank of New York, Securities and Exchange Commission, and Commodity Futures Trading Commission.

Kalman, R. E., 1960, A new approach to linear filtering and prediction problems, *Journal of Basic Engineering* 82, 35–45.

Kreiss, H.-O., 1962, Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren, *BIT Numerical Mathematics* 2, 153–181.

Kyle, A. S., 1985, Continuous auctions and insider trading, *Econometrica* 53, 1315–1335.

Le Cam, L., 1986, *Asymptotic Methods in Statistical Decision Theory*, Springer-Verlag.

MacKinnon, J. G., and M. D. Webb, 2018, The wild bootstrap for few (treated) clusters, *Econometrics Journal* 21, 114–135.

Menkveld, A. J., 2013, High frequency trading and the new market makers, *Journal of Financial Markets* 16, 712–740.

Newey, W. K., and K. D. West, 1987, A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix, *Econometrica* 55, 703–708.

Nielsen, C., 2021, `crypto-rl`: High-frequency limit order book data capture, GitHub repository (CC0 License).

Ntakaris, A., M. Magris, J. Kanniainen, M. Gabbouj, and A. Iosifidis, 2018, Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods, *Journal of Forecasting* 37, 852–866.

Pesaran, H. H., and Y. Shin, 1998, Generalized impulse response analysis in linear multivariate models, *Economics Letters* 58, 17–29.

Phillips, P. C. B., 1987, Time series regression with a unit root, *Econometrica* 55, 277–301.

Rigobon, R., 2003, Identification through heteroskedasticity, *Review of Economics and Statistics* 85, 777–792.

Spijker, M. N., 1991, On a conjecture by LeVeque and Trefethen related to the Kreiss matrix theorem, *BIT Numerical Mathematics* 31, 551–555.

Trefethen, L. N., and M. Embree, 2005, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*, Princeton University Press.

Vetterli, M., P. Marziliano, and T. Blu, 2002, Sampling signals with finite rate of innovation, *IEEE Transactions on Signal Processing* 50, 1417–1428. https://doi.org/10.1109/TSP.2002.1003065.

Wang, Y., D. Sornette, K. Wu, and S. C. Lera, 2024, Dynamic influence networks self-organize towards sub-critical financial instabilities, Swiss Finance Institute Research Paper 24-77. https://doi.org/10.2139/ssrn.4946318.

Binance, 2022a, Binance to exclude 13 BTC spot pairs in trading volume calculations, July 8. https://www.binance.com/en/support/announcement/detail/9541012e103c4ad3bfccbc54cd5f18ba.

Binance, 2022b, Binance launches ETH/BUSD zero-fee trading, August 23. https://www.binance.com/en/support/announcement/detail/5d8f9c0a68e24b82bff743a342dfa9b9.

Manole, T., and N. Ho, 2022, Refined convergence rates for maximum likelihood estimation under finite mixture models, Proceedings of Machine Learning Research 162, 14979–15006. https://proceedings.mlr.press/v162/manole22a.html.